Multivariate Moment Problems and the Gaussian Radon Transform

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ABSTRACT

We discuss various properties and problems related to the Gaussian Radon transform, particularly as applied to multivariate moment problems in finite dimensional Euclidean space. We propose a modification to a known method of bounded shape reconstruction from multivariate moments which can reconstruct both unbounded shapes and even functions square integrable with respect to a Gaussian weight. Related results include determinacy conditions for the multivariate and projection moments of a large class of functions, as well as explicit formulas for the Gaussian Radon transform of multivariate Hermite polynomials.

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Introduction

The primary goal of this dissertation is to establish that a method for the reconstruction of bounded shapes from their moments, described by Cuyt et al. in [3], can be extended to unbounded shapes. The proposed extension utilizes the Gaussian Radon transform, and the combination of these topics will bring with it some related problems. The shape reconstruction method falls in the intersection of two distinct fields of active research: multivariate moment problems and the Radon transform.

The task of determining a measure from the integrals of monomials is called a moment problem. While classical moment problems for positive measures on $\mathbb{R}$ have been well understood for at least a century, somewhat less is known in the corresponding study in multiple variables. The method presented by Cuyt et al. is a procedure which, given the moments of the indicator function $f(x)$ of a measurable set $A \subseteq \mathbb{R}^n$, reconstructs an approximation of A in the form of a pixel image (more precisely, an approximation to $f(x)$ restricted the nodes of a cubature formula). The initial focus of this dissertation will be a pivotal question of convergence in this paper, rather than the numerical method itself which hinges on the solution of a large and generally ill-determined linear system.

Alhough it is described as a shape reconstruction method, Cuyt et al. show that it can be applied more generally to positive measurable functions $f(x)$ with compact support. The method relies crucially on the approximation of the multivariate Stieltjes transform of $f(x)$ which is given by the integral representation

$$g(y) := \int_{\mathbb{R}^n} \frac{f(x)}{1 + \langle y, x \rangle} dx, \quad y \in \mathbb{R}^n.$$

When restricted to a one-dimensional subspace, the multivariate Stieltjes transform coincides with the univariate Stieltjes transform of the Radon transform of f in the direction of the subspace. That is, setting $y = z\omega$ for any $\omega \in S^{n-1}$ and $z \in \mathbb{R}$,

$$\int_{\mathbb{R}^n} \frac{f(x)}{1 + z\langle \omega, x \rangle} dx = \int_{\mathbb{R}} \frac{R_f(\omega, p)}{1 + zp} dp,$$

where $R_f : S^{n-1} \times \mathbb{R} \to \mathbb{R}$ is the Radon transform (RT) given by the integral of $f(x)$ over the hyperplane $\{x \in \mathbb{R}^n : \langle \omega, x \rangle = p\}$. Thus the problem is reduced that of approximating univariate Stieltjes transforms of the form

$$g_\omega(z) := \int_{\mathbb{R}} \frac{F(p)}{1 + zp} dp.$$

Note that $F(p) = R_f(\omega, p)$ also has compact support.

The $[M-1/M]$ Padé approximant to a power series $\sum_{k=0}^{\infty} c_k z^k$ is the rational function P/Q, where $P(z)$ and $Q(z)$ are polynomials of degree $M-1$ and M

respectively, satisfying

$$\frac{P(z)}{Q(z)} = \sum_{i=0}^{2M} c_k z^k + O(z^{2M+1}),$$

provided such polynomials exist. The series expansion of the Stieltjes transform is $\sum_{k=0}^{\infty} c_k(-z)^k$, where c_k are the moments of the function $F(p)$. If $F(p)$ has compact support then the Stieltjes transform is analytic in a neighborhood of zero and its $[M-1/M]$ Padé approximants converge locally uniformly in that neighborhood [1]. Since the moments of $F(p) = R_f(\omega, p)$ can be computed easily from the moments of $f(x)$, the shape reconstruction method method is theoretically justified.

Extending this method to unbounded shapes presents two substantial challenges. First, an unbounded set may not even have well defined moments, and the RT hyperplane integrals may be infinite. To this end, it is useful to introduce Gaussian weighted moments, and the Gaussian Radon transform (GRT) of f, given by

$$GR_f(\omega, p) = (2\pi)^{-\frac{n-1}{2}} \int_{\langle \omega, x\rangle = p} f(x) e^{-\frac{\|x-p\omega\|^2}{2}} dx. \tag{0.0.1}$$

The GRT as defined above was first studied in and of itself by Sengupta and Mihai in [9], although it showed up in the earlier work of Bogachev [**?**] for one example. Compared to the RT it has the advantages of a clear probabilistic interpretation, as well as being generalizable to infinite dimensional hilbert spaces [9]. For the purposes of this dissertation, it suffices to note that it can be applied to indicator functions of unbounded sets without fear of divergent hyperplane integrals.

Secondly, the Stieltjes transform of a function $F(p)$ of unbounded support may not be well defined over the real line. However, provided $F(p)$ has finite moments, the univariate Stieltjes transform can be defined as an analytic function on the half plane Im $z > 0$. Furthermore, under certain conditions, the $[M-1/M]$ Padé approximants constructed from the possibly formal series $\sum_{k=0}^{\infty} c_k(-z)^k$, will converge to the Stieltjes transform locally uniformly in the half plane [1]. If the function F is uniquely determined by its moments, then the Padé approximants converge as described. Hence, in chapters 2 and 3 we establish the following result:

0.0.1 Theorem. Let $f : \mathbb{R}^n \to [0, \infty)$ be a function with finite Gaussian weighted moments,

$$c_\alpha^G = (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} x^\alpha f(x) e^{-\frac{\|x\|^2}{2}} dx, \quad \alpha \in \mathbb{N}_0^n$$

If f is polynomially bounded, then the GRT of f defined by (0.0.1) is well defined for all $\omega \in S^{n-1}$, $p \in \mathbb{R}$. The $[M-1/M]$ Pade approximants computed from the moments $\{c_\alpha^G\}_{\alpha \in \mathbb{N}_0^n}$ converge locally uniformly in the upper half plane to the Stieltjes transform of $GR_f(\omega, \cdot)$ with a Gaussian weight,

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{GR_f(\omega, p)}{1 + zp} e^{-\frac{\|p\|^2}{2}} dp,$$

where Im $z > 0$.

Of particular interest to us are the cases when f is the indicator function of an unbounded measurable set $A \subseteq \mathbb{R}^n$ and when f is a polynomial. Unlike the RT, the GRT is well defined on the vector space of the polynomials on $\mathbb{R}^n$, raising the question of how exactly the GRT acts on polynomials. Some early computation indicates that there is a simple relationship between the Hermite polynomials and the GRT. For example in $\mathbb{R}^2$, the tensor product multivariate Hermite polynomials given by

$$H_{k,\ell}(x, y) := H_k(x)H_\ell(y), \quad x, y \in \mathbb{R}, \quad k, \ell \in \mathbb{N}_0,$$

where $\{H_k\}_{k\in\mathbb{N}_0}$ are the Hermite polynomials, appear to follow the relation,

$$GR_{H_{k,\ell}}(\omega, p) = H_{k+\ell}(p)\omega_1^k\omega_2^\ell, \quad k, \ell \in \mathbb{N}_0,$$

where $\omega = (\omega_1, \omega_2) \in S^1$. Extrapolating from the given computational evidence, in chapter 4 we are able to prove the following theorem, as well as an extension to the d-plane GRT:

0.0.2 Theorem. For multiindex $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$ let

$$H_\alpha(x) := H_{\alpha_1}(x_1)H_{\alpha_2}(x_2)\cdots H_{\alpha_n}(x_n), \quad x = (x_1 \ldots, x_n) \in \mathbb{R}^n,$$

where $\{H_k\}_{k\in\mathbb{N}_0}$ are the Hermite polynomials. Then for any $\omega = (\omega_1, \ldots, \omega_n) \in$

S^{n-1}, $p \in \mathbb{R}$ we have

$$GR_{H_\alpha}(\omega, p) = H_k(p)\omega^\alpha, \quad \alpha \in \mathbb{N}_0^n, \tag{0.0.2}$$

where $k = \alpha_1 + \cdots + \alpha_n$ and $\omega^\alpha = \omega_1^{\alpha_1} \cdots \omega_n^{\alpha_n}$.

Finally, the above claim gives a potential path toward solving a related problem. Sometimes from a modern scientific perspective, discrete models are prefered to continuous models. In particular, discretization is often necessary for computational methods such as the shape reconstruction procedure. Various discrete analogs to the Radon transform have been studied. See for example Beylkin's discrete RT defined over a finite two-dimensional grid [**?**], and Abouelaz and Ihsane's over the integer lattice $\mathbb{Z}^n$ [**?**]. A discrete GRT, on the other hand, has yet to be introduced. Discretizing formula (0.0.2) may fill this gap.

The Krawtchouk polynomials are orthogonal polynomials with respect to the binomial distribution. As the binomial distribution is a discrete version the Gaussian, so the Krawtchouk polynomials are discrete versions of the Hermite polynomials. Furthermore by a certain limiting procedure, the binomial distribution and Krawtchouk polynomials converge respectively to the Gaussian distribution and Hermite polynomials.

It seems reasonable to expect that some discrete GRT may be defined with a binomial — rather than Gaussian — weight by replacing the Hermite polynomials with the Krawtchouk in formula (0.0.2). This formal substitution allows us to define a new transform with a similar relation to the Krawtchouk polynomials. In chapter **??**, we introduce the descretization of formula (0.0.2) via the Krawtchouk polynomials, explore its properties, and discuss the relationship to known discrete RTs.

Chapter 1

Background

1.1 Basic Notation

First recall the conventional multi-index notation. Let $\mathbb{N}_0$ denote the nonnegative integers. A multi-index $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n) \in \mathbb{N}_0^n$ is an n-tuple of nonnegative integers. The degree of a multi-index is $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$. Multivariate exponentiation is defined as follows. For $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$,

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}.$$

Here we will use the convention $0^0 = 1$ so that for example $(x, y, z)^{(0,1,0)} = y$. We will use $e_i \in \mathbb{R}^n$ to represent the canonical unit vector

$$e_i = (0, \ldots, 0, 1, 0, \ldots, 0)$$

which can also be interpreted as a multi-index so for example

$$(x, y, z)^{2e_1 + e_3} = x^2 z$$

The multinomial formula gives a convenient expansion for multinomial powers.

Let $k \in \mathbb{N}_0$. Then

$$(b_1 + b_2 + \cdots + b_n)^k = \sum_{|\alpha|=k} \binom{k}{\alpha} b^\alpha$$

where the multinomial coefficients are defined

$$\binom{k}{\alpha} = \frac{k!}{\alpha_1!\alpha_2!\cdots\alpha_n!}.$$

Note that the multinomial expansion sums over all multi-indices α of degree k.

We denote the standard euclidean inner product

$$\langle x, y\rangle = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$

where $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$ and the Euclidean norm

$$\|x\| = \langle x, x\rangle = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

We may slightly abuse notation by using the inner product and norm in different dimensions, even in the same equation. This can be excused if one views each euclidean space $\mathbb{R}^n$ as embedded in the sequence space $\ell^2(\mathbb{R}^\mathbb{N})$ in which the inner product and norm are equivalent.

When discussing hyperplanes in $\mathbb{R}^n$ we index them by a unit normal vector, $\omega \in S^{n-1}$, and distance from origin $-\infty < p < \infty$, and we write for example the implicit hyperplane equation $\langle x, \omega\rangle = p$. Note that $\langle x, -\omega\rangle = -p$ represents the same hyperplane. It is perhaps more correct, in later defining the Radon and Gaussian Radon transforms, to identify these indexes and define the transforms over a projective space. We omit this discussion as it is not relevant within the scope of this work.

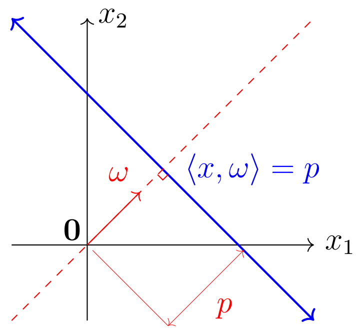

Figure 1.1.1: Hyperplane equation

Unless otherwise indicated all measures are Euclidean measures — that is the Borel measure associated with the standard Euclidean metric on a given region of integration. In particular the Euclidean measure on a hyperplane of $\mathbb{R}^n$ is equivalent to the Euclidean measure on $\mathbb{R}^{n-1}$. We choose for convenience to denote measures by their associated variable such as dx, dp, dz, and others. It should be stated that this notation, while uniform, is context dependent. For example in the integrals

$$\int_{\mathbb{R}^n} dx \qquad \text{and} \qquad \int_{\langle x,\omega\rangle = p} dx,$$

the measure dx is to be understood as the n-dimensional and $(n-1)$-dimensional Euclidean measure respectively.

For $1 \leq p < \infty$, $L^p = L^p(\mathbb{R}^n)$ is the space of Lebesgue measurable functions $f : \mathbb{R}^n \to \mathbb{R}$ satisfying

$$\int_{\mathbb{R}^n} |f(x)|^p \, dx < \infty.$$

As usual these functions are only defined modulo differences on sets of Lebesgue measure zero. For the Gaussian weight γ_n we similarly define $L^p(\gamma_n)$ for functions satisfying

$$\int_{\mathbb{R}^n} |f(x)|^p \gamma_n(x) \, dx < \infty$$

modulo Lebesgue measure zero sets.

1.2 Classical and multivariate moment problems

Let μ be a Borel measure on $\mathbb{R}$. For $k \in \mathbb{N}_0$, define the kth moment of μ as

$$c_k = \int_{-\infty}^{\infty} x^k \, d\mu(x)$$

provided the integral converges. The sequence $(c_k)_{k \in \mathbb{N}_0}$ is called a moment sequence or a moment problem, and the measure μ is called a solution to the moment problem. Loosely speaking a moment problem poses the question: Under given constraints (e.g. measures supported within a fixed subset of $\mathbb{R}^n$), to what extent can one determine a solution μ (if one exists) from a moment sequence?

The classical study of moment problems is divided into three cases depending on the support of μ. In each case there is a standard choice of support, from which general results are often derived by a change of variables.

Markov (or **Haussdorf**) moment problems within a bounded support.

$$\text{supp}(\mu) \subseteq [0, 1]$$

Stieltjes moment problems within a one-sided infinite support.

$$\text{supp}(\mu) \subseteq [0, \infty)$$

Hamburger moment problems within a two-sided infinite support.

$$\text{supp}(\mu) \subseteq \mathbb{R}$$

In some special cases we will denote the moments c_k in a particular form:

(i) When μ is absolutely continuous with respect to the Lebesgue measure dx, moments can be defined by

$$c_k = \int f(x) x^k dx$$

where $f(x)$ is the density of μ with respect to dx.

(ii) When μ is the Lebesgue measure on a set $A \subset \mathbb{R}$ of finite measure, moments can be defined by

$$c_k = \int_A x^k dx.$$

We often use these representations interchangeably. For example, we may say a function f is determinate, or a set A is indeterminate, if the corresponding measure μ has that property.

There are two natural questions one can ask about a moment problem:

Solvability: Does a solution μ exist possessing the given moments?

Determinacy: Is a solution unique? If not, what can be said about the set of solutions?

In the context of the shape reconstruction method of Chapter 3 it is taken as a given that a solution exists, but that we have access only to the moments. Furthermore, in practical application we encounter a followup question to determinacy: How can we reconstruct the solution?

In the classical cases (Markov, Stieltjes, Hamburger) these questions have been more or less resolved. Precise conditions for solvable and determinate moment problems have been found, and the nature of solution sets to indeterminate moment problems are well understood. A good reference for the theory of classical moment problems can be found in [16].

For instance, all solvable Markov moment problems are unique. This result follows from the Weierstrass approximation theorem:

1.2.1 Proposition. All Markov moment problems are determinate. If μ, ν are Borel measures on the unit interval $[0,1]$ with equal moments,

$$\int_0^1 x^k d\mu = \int_0^1 x^k d\nu, \qquad k \in \mathbb{N}_0,$$

then $\mu = \nu$.

We now define multivariate moment problems. Let μ be a Borel measure, now defined on $\mathbb{R}^n$. For $\alpha \in \mathbb{N}_0^n$. Define the αth multivariate moment of f as

$$c_\alpha = \int_{\mathbb{R}^n} x^\alpha \ d\mu(x).$$

1.2.2 Example. Let $A = [0,1]^n \subseteq \mathbb{R}^n$ be the unit cube and μ the Lebesgue measure on A. Then μ has multivariate moments

$$\begin{aligned}
\int_{\mathbb{R}^n} x^\alpha \ d\mu(x) &= \int_{[0,1]^n} x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} dx \\
&= \int_0^1 x_1^{\alpha_1} \ dx_1 \int_0^1 x_2^{\alpha_2} \ dx_2 \cdots \int_0^1 x_n^{\alpha_n} \ dx_n \\
&= \frac{1}{(\alpha_1+1)(\alpha_2+1)\cdots(\alpha_n+1)}
\end{aligned}$$

Similarly the Lebesgue measure μ on any rectanglular region $A = \prod_{i=1}^n [a_i, b_i]$ has moments

$$\int_A x^\alpha \, d\mu(x) = \prod_{i=1}^{n} \int_{a_i}^{b_i} x_i^{\alpha_i} \, dx_i = \prod_{i=1}^{n} \frac{b_i^{\alpha_i+1} - a_i^{\alpha_i+1}}{\alpha_i + 1}$$

The moment sequence $(c_\alpha)_{\alpha\in\mathbb{N}_0^n}$ is now multi-indexed, but the notions of solvability and determinacy are analogous: It is solvable provided there exists a Borel measure with these moments, and it is determinate provided such a measure is unique. Here, we divide multivariate moment problems into two cases:

Bounded multivariate moment problems within a bounded support.

$$\operatorname{supp}(\mu) \subseteq [0,1]^n$$

Unbounded multivariate moment problems within an unbounded support.

$$\operatorname{supp}(\mu) \subseteq \mathbb{R}^n$$

While conditions for the solvability or determinacy of multivariate moment problems have been found in certain cases, they are nowhere near as comprehensive as those of classical moment problems. One straightforward, albeit limited, approach is to reduce and n-variate moment problem to a set of n univariate moment problems [10]. For example, we employ this strategy in chapter 2.

As in the classical case of Markov moment problems, determinacy is a consequence of the Stone-Weierstrass theorem for n-variate polynomials:

1.2.3 Proposition. If μ is a Borel measure on the unit cube $A = [0,1]^n$, with moments

$$c_\alpha = \int_A x^\alpha d\mu, \qquad \alpha \in \mathbb{N}_0^n$$

then the moment problem $(c_\alpha)_{\alpha \in \mathbb{N}_0^n}$ is determinate.

We present a simple alternate proof in chapter 2 via the Radon transform.

1.3 Gaussian measures and the Hermite polynomials

We review the definitions of univariate and multivariate Gaussian measures and Hermite polynomials, and discuss some basic properties. An extensive exposition of Gaussian measures can be found in [2]. The univariate Hermite polynomials are well known, but there are multiple multivariate generalizations. We use a simple product definition.

First we review some basic properties of the univariate Gaussian measure.

1.3.1 Definition. Let $\gamma(p) : \mathbb{R} \to \mathbb{R}$ be the standard Gaussian density on $\mathbb{R}$,

$$\gamma(p) = \frac{1}{\sqrt{2\pi}} e^{\frac{-p^2}{2}}$$

and $\gamma_n(x) : \mathbb{R}^n \to \mathbb{R}$ the standard Gaussian density on $\mathbb{R}^n$,

$$\gamma_n(x) = (2\pi)^{-\frac{n}{2}} e^{\frac{-\|x\|^2}{2}}.$$

Note that $\gamma = \gamma_1$.

The normalization constant $(2\pi)^{-\frac{n}{2}}$ ensures that $\gamma_n(x)$ defines a probability distribution,

$$\int_{-\infty}^{\infty} \gamma(p)\ dp = 1.$$

The Gaussian density is analytic and satisfies

$$\gamma'(p) = -p\gamma(p)$$

in the univariate case and more generally,

$$\frac{\partial}{\partial x_i}\gamma_n(x) = -x_i\gamma_n(x), \qquad i = 1, \ldots, n.$$

Notably, the exponential decay of $\gamma_n(x)$ at infinity means that the Gaussian density dominates polynomial growth in the sense that $P(x)\gamma_n(x) \to 0$ as $\|x\| \to \infty$, even

$$\int_{\mathbb{R}^n} P(x)\gamma_n(x)\ dx < \infty$$

for all polynomials $P : \mathbb{R}^n \to \mathbb{R}$.

1.3.2 Definition. The *Gaussian moments* of a function $f : \mathbb{R}^n \to \mathbb{R}$ are,

$$c_\alpha^\gamma = \int_{\mathbb{R}^n} f(x)x^\alpha\gamma(x)\ dx$$

for $\alpha \in \mathbb{N}_0^n$, provided they exist.

1.3.3 Example. Let $f : \mathbb{R} \to \mathbb{R}$ be the constant function $f(p) = 1$. The Gaussian moments of f, i.e. the moments of $\gamma(p)$, are

$$c_k^\gamma = \begin{cases} (k-1)!!, & k \text{ even} \\ 0, & k \text{ odd} \end{cases}$$

We have already seen $c_0 = 1$. Furthermore it is not hard to see that for odd k,

$$c_k^\gamma = \int_{-\infty}^{\infty} p^k \gamma(p)\ dp = 0$$

since $p^k\gamma(p)$ is an odd function.

Now for $\ell = 1, 2, \ldots$, we derive a recurrence formula for $c_{2\ell}$ by integrating by parts

$$\begin{aligned} c_{2\ell}^\gamma &= \int_{-\infty}^{\infty} p^{2\ell}\gamma(p)\ dp \\ &= -\int_{-\infty}^{\infty} p^{2\ell-1}\left(-p\gamma(p)\right)\ dp \\ &= -\left[p^{2\ell-1}\gamma(p)\right]_{-\infty}^{\infty} + (2\ell-1)\int_{-\infty}^{\infty} p^{2\ell-2}\gamma(p)\ dp \\ &= (2\ell-1)c_{2\ell-2}^\gamma. \end{aligned}$$

Thus by induction,

$$c_{2\ell}^\gamma = (2\ell-1)(2\ell-3)\cdots c_0 = (2\ell-1)!!.$$

As an alternate derivation, for readers familiar with the gamma function $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$ we present the following: Since $p^{2\ell}\gamma(p)$ is even,

$$\int_{-\infty}^{\infty} p^{2\ell}e^{-\frac{p^2}{2}}dp = 2\int_0^\infty p^{2\ell}e^{-\frac{p^2}{2}}dp$$

Now substituting $t = \frac{p^2}{2}$, whereby $dt = p\ dp$,

$$\begin{aligned} 2\int_0^\infty p^{2\ell}e^{-\frac{p^2}{2}}dp &= 2^{\ell+\frac{1}{2}}\int_0^\infty t^{\ell-\frac{1}{2}}e^{-t}dt \\ &= 2^{\ell+\frac{1}{2}}\Gamma\left(\ell+\frac{1}{2}\right) \end{aligned}$$

so that

$$c_{2\ell}^{\gamma} = \frac{2^\ell}{\sqrt{\pi}}\Gamma\left(\ell+\frac{1}{2}\right) = (2\ell-1)!!$$

A relationship between the univariate and multivariate Gaussian densities can be seen as follows: Let $x, y \in \mathbb{R}^n$, $n \geq 2$. Suppose $y = p\omega$ where $\omega \in S^{n-1}$ and $p \in \mathbb{R}$, so that $p\omega$ is the orthogonal projection of x onto the span of y. Then the Pythagorean relation,

$$\|x\|^2 = \|x - p\omega\|^2 + \|p\omega\|^2 = \|x - p\omega\|^2 + p^2$$

implies that

$$\begin{aligned}\gamma_n(x) &= (2\pi)^{-\frac{n}{2}} e^{\frac{-\|x\|^2}{2}} \\ &= (2\pi)^{-\frac{n-1}{2}} e^{\frac{-\|x-p\omega\|^2}{2}} (2\pi)^{-\frac{1}{2}} e^{\frac{-p^2}{2}} \\ &= \gamma_{n-1}(x-p\omega)\gamma(p)\end{aligned}$$

The equation

$$\gamma_n(x) = \gamma_{n-1}(x-p\omega)\gamma(p) \tag{1.3.1}$$

is in some respect the defining property of the Gaussian measure described below. Indeed if $x = (x_1, x_2, \ldots, x_n)$, by repeated application of (1.3.1) one can write the decomposition

$$\gamma_n(x) = \prod_{k=1}^{n} \gamma(x_k).$$

Thus the $\gamma_n(x)$ is the product of n copies of γ. The standard Gaussian measure γ^n is thus a measure whose cardinal projections are standard Gaussian measures.

1.3.4 Example. Let $f : \mathbb{R}^n \to \mathbb{R}$ be the constant function $f(x) = 1$. The

Gaussian moments of f are the multivariate moments of $\gamma_n(x)$, and can be written

$$\begin{aligned} c_\alpha^\gamma &= \int_{\mathbb{R}^n} x^\alpha \gamma_n(x)\ dx \\ &= \prod_{i=1}^n \int_{-\infty}^{\infty} x_i^{\alpha_i} \gamma(x_i)\ dx \\ &= \prod_{i=1}^n c_{\alpha_i}^\gamma \end{aligned}$$

where $\{c_k^\gamma\}_{k\in\mathbb{N}}$ are the univariate moments of $\gamma(p)$. Thus $c_\alpha^\gamma = 0$ if any α_i is odd. Otherwise, if every α_i is even then

$$c_\alpha^\gamma = \prod_{i=1}^n (\alpha_i - 1)!!$$

1.3.5 Definition. The **Hermite polynomials** can be defined by the Rodrigues formula

$$H_k(p) = (-1)^k \frac{\gamma^{(k)}(p)}{\gamma(p)}$$

We can derive a recurrence relation for H_k with some basic differential calculus. First multiplying both sides by $(-1)^k\gamma(p)$, we get

$$\gamma^{(k)}(p) = (-1)^k \gamma(p) H_k(p)$$

Now we differentiate both sides

$$\gamma^{(k+1)}(p) = (-1)^k(-p\gamma(p)H_k(p) + \gamma(p)H_k'(p)) = (-1)^{k+1}\gamma(p)(pH_k(p) - H_k'(p))$$

so that, dividing both sides by $(-1)^{k+1}\gamma(p)$, we have

$$H_{k+1}(p) = pH_k(p) - H_k'(p)$$

Since $H_0(p)$ is trivially the constant 1 we compute the first handful of polynomials:

$$H_0(p) = 1$$

$$H_1(p) = p$$

$$H_2(p) = p^2 - 1$$

$$H_3(p) = p^3 - 3p$$

$$H_4(p) = p^4 - 6p^2 + 3$$

$$H_5(p) = p^5 - 10p^3 + 15p$$

Note that each H_k is a monic polynomial of degree k, and H_k is an even (odd resp.) function if k is even (odd resp.). The Hermite polynomials are mutually orthogonal in the sense that for nonnegative integers $\ell < k$,

$$\int_{-\infty}^{\infty} H_\ell(p)H_k(p)\gamma(p)\ dp = 0 \tag{1.3.2}$$

To see this, substitute the Rodrigues formula for H_k,

$$(-1)^k \int_{-\infty}^{\infty} H_\ell(p)\gamma^{(k)}(p)\ dp$$

and integrate by parts

$$(-1)^k \left(\left[H_k(p)\gamma^{(k-1)}(p)\right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} H_\ell'(p)\gamma^{(k-1)}(p)\ dp \right)$$
$$= (-1)^{k-1} \int_{-\infty}^{\infty} H_\ell'(p)\gamma^{(k-1)}(p)$$

where the boundary term $\left[H_k(p)\gamma^{(k-1)}(p)\right]_{-\infty}^{\infty}$ vanishes because of $\gamma(p)$'s rapid decay at $\pm\infty$. Repeat this process until H_ℓ is annihilated.

The Hermite polynomials form a complete orthogonal basis for $L^2(\gamma)$, the space of function $f : \mathbb{R} \to \mathbb{R}$ for which $\int_{-\infty}^{\infty} f(p)^2\gamma(p)\ dp < \infty$. This is a Hilbert space with inner product $\langle f, g\rangle = \int_{-\infty}^{\infty} |f(p)g(p)|\gamma(p)\ dp$.

1.3.6 Proposition. The linear span of the Hermite polynomials $H_0(p), H_1(p), \ldots$ is dense in $L^2(\mathbb{R}, \gamma)$.

1.3.7 Corollary. For any positive function $f \in L^2(\gamma)$ the moment problem given by

$$c_n^\gamma = \int_{-\infty}^{\infty} f(p)\gamma(p)dp$$

is determinate.

In the chapter 4 we will make use of a different definition for the Hermite polynomials, as Taylor series coefficients for an exponential generating function.

1.3.8 Lemma. The Hermite polynomials have the **generating function**,

$$\phi(p) = e^{pt-\frac{t^2}{2}} = \sum_{k=0}^{\infty} \frac{H_k(p)}{k!} t^k.$$

which converges absolutely for all $p, t \in \mathbb{R}$.

Proof. Note that

$$e^{pt-\frac{t^2}{2}} = \frac{e^{-\frac{(p-t)^2}{2}}}{e^{-\frac{p^2}{2}}} = \frac{\gamma(p-t)}{\gamma(p)}$$

so that the Taylor coefficients are precisely

$$\left[\frac{d^k}{dt^k} \frac{\gamma(p-t)}{\gamma(p)}\right]_{t=0} = (-1)^k \frac{\gamma^{(k)}(p)}{\gamma(p)} = H_k(p).$$

□

We can now define a straight forward multivariate analogue or the Hermite polynomials

1.3.9 Definition. For $\alpha \in \mathbb{N}_0^n$ we define the **multivariate Hermite polynomial** H_α by

$$H_\alpha(x) = H_{\alpha_1}(x_1) \cdots H_{\alpha_n}(x_n), \qquad x = (x_1, \ldots x_n) \in \mathbb{R}^n$$

The multivariate Hermite polynomials have many useful properties corresponding to those of the classical polynomials. In particular we have the Rodrigues formula

$$H_\alpha(x) = (-1)^{|\alpha|} \frac{\partial_\alpha \gamma_n(x)}{\gamma_n(x)}$$

where $\partial_\alpha = \frac{\partial^{\alpha_1+\cdots+\alpha_n}}{\partial x_1^{\alpha_1}\cdots\partial x_n^{\alpha_n}}$ is a partial differential operator. Just like the classical case, this formula can be used to define H_α as multivariate Taylor coefficients for the generating function:

$$\phi(x) = e^{\langle x,y\rangle - \frac{\|y\|^2}{2}} = \frac{\gamma_n(x-y)}{\gamma_n(x)} = \sum_{\alpha\in\mathbb{N}_0^n} \frac{H_\alpha(x)}{\alpha!} y^\alpha$$

which we note is absolutely convergent for all $x, y \in \mathbb{R}^n$. Here we use the same symbol ϕ for the univariate and multivariate generating functions at the risk of confusion. It should be clear from context.

1.4 The Radon and Gaussian Radon transforms

Let f be a function on the n-dimensional Euclidean space $\mathbb{R}^n$. We imagine taking "slices" of f by restricting it to a $(n-1)$-dimensional hyperplane Λ. These hyperplanes form the domain of the Radon Transform. More precisely, the Radon Transform associates each slice with a corresponding integral

$$R_f(\Lambda) = \int_\Lambda f(x)\, dx,$$

which can be thought of as a $(n-1)$-dimensional measurement of f. To be more precise we parametrize the collection of hyperplanes Λ by normal vector, $\omega \in$

S^{n-1}, and (signed) distance from the origin $-\infty < p < \infty$. Indeed any hyperplane can be described in the form $\Lambda = \{x \in \mathbb{R}^n : \langle x, \omega \rangle = p\}$. As mentioned, there is a slight inconsistency in these definitions where a hyperplane Λ can be indexed by both $\langle \omega, x \rangle = p$ and $\langle -\omega, x \rangle = -p$. This difference is inconsequential for our purposes so we choose the latter definition for clarity.

The Radon transform [6] (RT) is a thing that I will discuss the history of, with references, in this paragraph. The transform gets its name from Johann Radon, whose first defined the transform in the form below in 1917, although a similar transform was introduced by Paul Funk in 1911. It is interesting to note that the primary application of the RT in medical imaging (CT scans) was not invented for another half century. I can only hope that in 2076 my dissertation will serve as an absorbent coffee coaster for a sleep deprived student.

1.4.1 Definition. Let f be a nonnegative measurable function on $\mathbb{R}^n$. The **Radon transform** $R_f : S^{n-1} \times \mathbb{R} \to \mathbb{R}$ of f is a function which, given a unit vector $\omega \in \mathbb{R}^n$ and $p \in \mathbb{R}$, is defined as

$$R_f(\omega, p) = \int\limits_{\langle x, \omega \rangle = p} f(x)\, dx,$$

provided the integral converges.

1.4.2 Example. For computation it is convenient to write the Radon Transform with an explicit isometric parameterization $x(t)$ of the hyperplane $\langle x, \omega \rangle = p$. In particular we note that there exists a map $x : \mathbb{R}^{n-1} \to \mathbb{R}^n$ such that $x(0) = p\omega$

and

$$R_f(\omega, p) = \int_{\mathbb{R}^{n-1}} f(x(t))dt.$$

For reference let's specify an hyperplane parameterization for the $n = 2$ case. In $\mathbb{R}^2$ often we identify ω with the angle $0 \leq \theta < 2\pi$ such that $\omega = (\cos\theta, \sin\theta)$. We define $x(t)$ by

$$x(t) = (t\sin\theta + p\cos\theta, -t\cos\theta + p\sin\theta), \qquad -\infty < t < \infty.$$

It is not difficult to show that $x(0) = p\omega$, $\langle x(t), \omega\rangle = p$, and most importantly

$$R_f(\omega, p) = \int_{-\infty}^{\infty} f(t\sin\theta + p\cos\theta, -t\cos\theta + p\sin\theta)\ dt$$

1.4.3 Example. Let $B(r) = \{x : |x| \leq r\} \subseteq \mathbb{R}^n$ be the ball of radius r.

$$R_{B(r)}(\omega, p) = \begin{cases} V_{n-1}\left(\sqrt{r^2 - p^2}\right), & |p| \leq r \\ 0 & |p| > r \end{cases}$$

where $V_n(r)$ is the volume of a hypersphere of radius r,

$$V_n(r) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} r^n.$$

Note that as a bounded domain, we can guarantee $B(r)$ is integrable on all hyperplanes. In fact, the RT of any subset $A \subseteq B(r)$ is bounded by

$$R_A \leq V_{n-1}(r).$$

Also note that $B(r)$ is rotation invariant, hence $R_{B(r)}(\omega, p)$ is independent of ω.

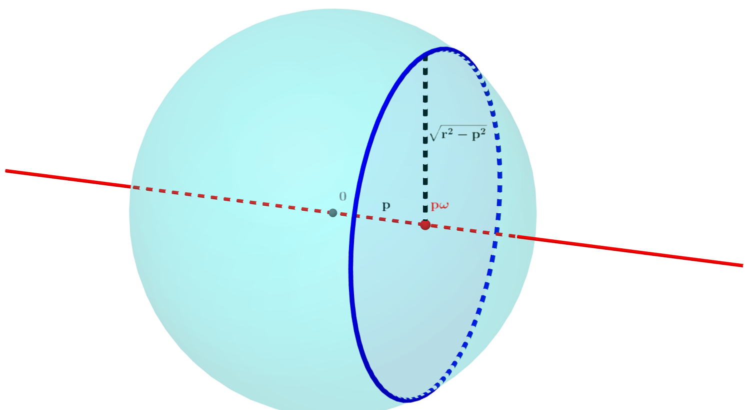

Figure 1.4.1: The RT of a ball

We can use this formula to determine the RT of an annulus. Let $A(r_1, r_2) = \{x \in \mathbb{R}^n : r_1 \leq \|x\| \leq r_2\}$. Then

$$\begin{aligned} R_{A(r_1,r_2)}(\omega, p) &= R_{B(r_2)}(\omega, p) - R_{B(r_1)}(\omega, p) \\ &= \begin{cases} V_{n-1}(\sqrt{r_2^2 - p^2}) - V_{n-1}(\sqrt{r_1^2 - p^2}), & |p| < r_1 \\ V_{n-1}(\sqrt{r_2^2 - p^2}), & r_1 \leq |p| \leq r_2 \\ 0, & |p| \geq r_2 \end{cases} \end{aligned}$$

Now consider an example of an unbounded region.

1.4.4 Example. Let $S = \{(x, y) : |y| \leq \frac{1}{2}\} \subseteq \mathbb{R}^2$ be a strip centered on the x-axis with width 1. Clearly if $\theta = \pi/2$ or $3\pi/2$ then

$$R_S(\theta, p) = \begin{cases} \infty, & |p| \leq \frac{1}{2} \\ 0, & |p| > \frac{1}{2} \end{cases}.$$

Otherwise,

$$R_S(\theta, p) = \sec\theta$$

Note because S is unbounded, that R_S is not only unbounded, but even divergent in some cases.

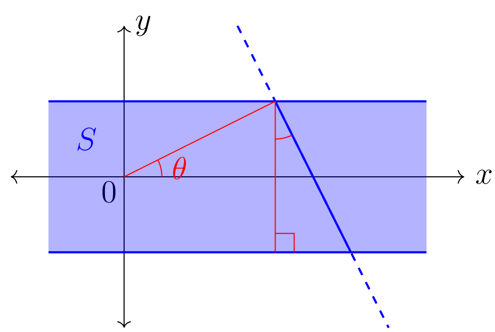

Figure 1.4.2: RT of a strip

Our main addition to previous work will be the use of a modified RT, the Gaussian Radon transform (GRT) [9] [7] [8]. This transform is very similar to the RT, but the inclusion of a Gaussian density $\gamma_{n-1}(x)$ in the integral allows for convergence on a larger class of functions f. This includes for example unbounded regions. In a broader context the Gaussian Radon transform also has the advantages of generalizing to infinite dimensional Hilbert spaces [8] (on which the Lebesgue measure is not defined), as well as having a natural probabilistic interpretation.

1.4.5 Definition. The *Gaussian Radon transform* $GR_f : S^{n-1} \times \mathbb{R} \to \mathbb{R}$ of f is defined similarly to the RT. Given $\omega \in S^{n-1}$ and $-\infty < p < \infty$, the GRT is

$$GR_f(\omega, p) = \int_{\langle x,\omega\rangle = p} f(x)\gamma_{n-1}(x - p\omega)\; dx.$$

provided the integral converges. Note the Gaussian density

$$\gamma_{n-1}(x - p\omega) = (2\pi)^{-(n-1)/2} e^{-\|x-p\omega\|^2/2}$$

is centered on the point of the hyperplane closest to the origin.

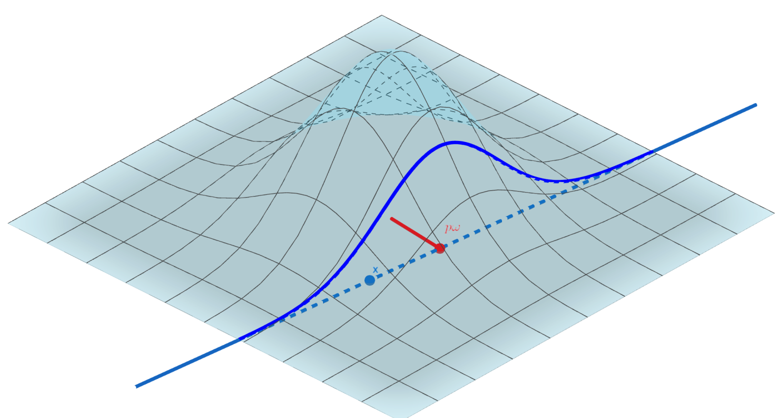

Figure 1.4.3: The GRT

1.4.6 Remark. It may be helpful to understand the GRT as a simple modification of the RT with respect to a Gaussian measure on $\mathbb{R}^n$. From the relation

$$\int\limits_{\langle x,\omega\rangle=p} f(x)\gamma_n(x)\ dx = \int\limits_{\langle x,\omega\rangle=p} f(x)\gamma_{n-1}(x)\ dx\ \gamma(p),$$

we can express the GRT of f in terms of the RT of the function $g(x) = f(x)\gamma_n(x)$:

$$R_g(\omega, p) = GR_f(\omega, p)\gamma(p), \qquad g(x) := f(x)\gamma_n(x). \tag{1.4.1}$$

The relation above provides decent intuition for the GRT, and is also a useful tool proving some basic properties of the transform. Expand on intuition (maybe with example.)

1.4.7 Example. If $x(t) : \mathbb{R}^{n-1} \to \mathbb{R}^n$ is a parametrization of $\langle x, \omega\rangle = p$ as described above, then

$$GR_f(\omega, p) = \int_{\mathbb{R}^{n-1}} f(x(t))\gamma_{n-1}(t)dt$$

In particular for $f : \mathbb{R}^2 \to \mathbb{R}$

$$GR_f(\omega, p) = \int_{-\infty}^{\infty} f(t\sin\theta + p\cos\theta, -t\cos\theta + p\sin\theta)\gamma(t)\ dt$$

where $\omega = (\cos\theta, \sin\theta)$.

1.4.8 Example. The Gaussian Radon transform is bounded for any measurable region $A \subseteq \mathbb{R}^n$. Indeed

$$GR_A(\omega, p) \le \int\limits_{\langle x,\omega\rangle=p} \gamma_{n-1}(x)\ dx = 1$$

Now, imagine sweeping a hyperplanar "slice" across $\mathbb{R}^n$. As a function of p, $R(\omega, p)$ can be seen as a projection of f onto the linear subspace spanned by ω. Intuitively, since the parallel slices covere the entirety of the space with no overlap, it is not surprising that integrating this projection over $-\infty < p < \infty$ we get the same result as the n-fold integral of f over $\mathbb{R}^n$.

$$\int_{-\infty}^{\infty} R(\omega, p)\ dp = \int_{\mathbb{R}^n} f(x)\ dx$$

The following theorem further generalizes this observation:

1.4.9 Proposition (Slice Theorem)**.** If $f : \mathbb{R}^n \to \mathbb{R}$ and $F : \mathbb{R} \to \mathbb{R}$ are measureable, then

$$\int_{-\infty}^{\infty} R_f(\omega, p) F(p)\ dp = \int_{\mathbb{R}^n} f(x) F(\langle x, \omega \rangle)\ dx, \tag{1.4.2}$$

provided

$$\int_{\mathbb{R}^n} f(x) F(\langle x, \omega \rangle)\ dx < \infty.$$

In particular, the slice relation holds under the following conditions:

(i) $f \in L^1(\mathbb{R})$ and F is bounded,

(ii) $f \in L^1(\mathbb{R})$ has compact support and F is locally bounded.

Proof. Pulling $F(p)$ into the RT integral, we can substitute $p = \langle x, \omega \rangle$,

$$\int_{-\infty}^{\infty} R_f(\omega, p) F(p)\ dp = \int_{-\infty}^{\infty} \int_{\langle x,\omega\rangle = p} f(x) F(\langle x, \omega \rangle)\ dx\ dp$$

then it remains to show that the itterated integrals can be combined. This is a relatively straightforward application of Fubini's theorem. Assume for example $\omega = (0, \ldots, 0, 1)$, so that $\langle x, \omega \rangle = x_n$ for any $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. In this simple case Fubini's theorem says for any measurable function $g : \mathbb{R}^n \to \mathbb{C}$

$$\int_{-\infty}^{\infty} \int_{\mathbb{R}^{n-1}} g(x_1, \ldots, x_{n-1}, p)\ dx_1 \cdots dx_{n-1}\ dp = \int_{\mathbb{R}^n} g(x_1, \ldots, x_n) dx$$

provided that either integral converges. The general direction $\omega \in S^{n-1}$ is equivalent to this special case after a rotation (under which the Lebesgue integrals are invariant).

Fubini's theorem only requires that the left hand integral converges. We show this condition is satisfied in the two cases

(i) Since F is bounded, say $|F| \leq M$ for some $M > 0$. We have

$$\int_{\mathbb{R}^n} |f(x) F(\langle x, \omega \rangle)|\ dx \leq \int_{\mathbb{R}^n} |f(x)| M\ dx = M \int_{\mathbb{R}^n} |f(x)|\ dx < \infty$$

since $f \in L^1(\mathbb{R}^n)$.

(ii) If $\mathrm{supp}(f)$ is compact then so is $\{p : \langle x, \omega \rangle = p \text{ for some } x \in \mathrm{supp}(f)\}$, since the mapping $x \mapsto \langle x, \omega \rangle$ is continuous. Thus $F(\langle x, \omega \rangle)$ is bounded on supp(f) and the rest of the proof is identical to the previous case.

$\square$

If $F(p) = e^{-ip}$ and $f(x)$ is such that $\int_{-\infty}^{\infty} R_f(\omega, p)dp < \infty$ then (1.4.2) becomes the well known Fourier slice theorem

$$\int_{-\infty}^{\infty} R_f(\omega, p)e^{-ip}\, dp = \int_{\mathbb{R}^n} f(x)e^{-i\langle x,\omega\rangle}\, dx,$$

which is often articulated as saying that the 1-dimensional Fourier transform of the Radon transform is the n-dimensional Fourier transform of f.

An early and natural question in the study of the RT is that of inversion. Radon himself derived the "Radon inversion formula" [11] [12], which is often proved via the above Fourier slice theorem. The groundbreaking inversion formula is the basis for what, in application, called "filtered back-propagation".

If one is interested in inverting the RT then a prerequisite concern is of course: Is the transform injective? The answer clearly depends on what space we draw the function f from. Radon [11] [12] provides a set of sufficient regularity conditions such that the RT is invertible. On the other hand counterexamples have been constructed by, for example, Lawrence Zalcman [15], of continuous and nontrivial functions for which the RT is identically zero.

By way of the relation (1.4.1) we can prove an analogous slice theorem for the GRT. Mihai and Sengupta give a more general result for the GRT on real, separable, infinite Hilbert spaces [9].

1.4.10 Proposition (Gaussian Slice Theorem)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ and $F : \mathbb{R} \to \mathbb{R}$ be measurable functions. If $f \in L^1(\mathbb{R}^n, \gamma_n)$ and F is bounded then

$$\int_{-\infty}^{\infty} GR_f(\omega, p)F(p)\gamma(p)\ dp = \int_{\mathbb{R}^n} f(x)F(\langle x, \omega\rangle)\gamma_n(x)\ dx. \tag{1.4.3}$$

1.4.11 Remark. Again the more general Fubini condition

$$\int_{\mathbb{R}^n} |f(x)F(\langle x, \omega\rangle)\gamma_n(x)|dx < \infty$$

may be used. Of course, whatever conditions on f and F are sufficient for convergence in (1.4.2) are then sufficient conditions to be checked of $f(x)\gamma_n(x)$ and $F(p)\gamma(p)$.

Proof. From (1.4.1)

$$\int_{-\infty}^{\infty} GR_f(\omega, p)F(p)\gamma(p)\ dp = \int_{-\infty}^{\infty} R_g(\omega, p)F(p)\ dp$$

where $g(x) = f(x)e^{-\|x\|^2/2}$. Then applying the slice theorem:

$$\int_{-\infty}^{\infty} R_g(\omega, p)F(p)\ dp = \int_{\mathbb{R}^n} f(x)F(\langle x, \omega\rangle)\gamma_n(x)\ dx,$$

completing the proof. □

1.5 The Radon transform of measures

Taking from the context of classical moment problems we should define the notion of the Radon transform of a measure μ. Let μ be a Borel measure on $\mathbb{R}^n$ with finite moments c_α, $\alpha \in \mathbb{N}_0^n$. The projection $\pi_\omega : \mathbb{R}^n \to \mathbb{R}$ given by

$$\pi_\omega(x) = \langle x, \omega \rangle$$

is a Borel function, thus we may define the push-forward measure $\mu_\omega = \mu \circ \pi_\omega^{-1}$ which for Borel sets $A \subseteq \mathbb{R}$ is given by,

$$\mu_\omega(A) = \mu(\pi_\omega^{-1}(A)) = \mu(\{x : \langle x, \omega \rangle \in A\}).$$

We may call μ_ω the **marginal projection measure** of μ with direction vector ω.

Notice that if μ is absolutely continuous with density representation $\mu = f(x)dx$ then the slice theorem with $F(p)$ the characteristic function of A gives,

$$\mu(\pi_\omega^{-1}(A)) = \int_{\langle x,\omega\rangle \in A} f(x)dx = \int_A R_f(\omega, p)dp$$

so that in fact μ_ω is absolutely continuous with respect to the Lebesgue measure dp and it's density is precisely the Radon transform $R_f(\omega, p)$. In this sense the marginal projection μ_ω is a natural generalization of the RT, so we define

1.5.1 Definition. The Radon transform of a Borel measure μ on $\mathbb{R}$ is defined as the push-forward measure $\mu_\omega = \mu \circ \pi_\omega^{-1}$ on $\mathbb{R}$, where $\pi_\omega : \mathbb{R}^n \to \mathbb{R}$ is the projection

$$\pi_\omega(x) = \langle x, \omega \rangle.$$

We may denote this measure by $R_\mu^\omega = \mu_\omega$.

1.5.2 Remark. The slice theorem is analogous to — in fact a special case of — the change of variables formula for the push-forward measure R_μ^ω:

$$\int_{-\infty}^{\infty} F(p) dR_\mu^\omega(p) = \int_{\mathbb{R}^n} F(\pi_\omega(x)) d\mu$$

where $F : \mathbb{R} \to \mathbb{R}$ is integrable with respect to dR_μ^ω if and only if $F \circ \pi_\omega : \mathbb{R}^n \to \mathbb{R}$ is integrable with respect to μ.

1.6 The Stieltjes transform

Let μ be a Borel measure with finite moments $c_k = \int_{-\infty}^{\infty} p^k d\mu$.

1.6.1 Definition. The function $g : \mathbb{C} \to \mathbb{C}$ defined by

$$g(z) = \int_{-\infty}^{\infty} \frac{d\mu}{1 + zp}$$

is called the Stieltjes transform of a Borel measure μ on $\mathbb{R}^n$.

It is worth noting at the outset that $g(z)$ is defined at least for non-real z. To see this we note that if Im $z \neq 0$,

$$\begin{aligned}\frac{1}{|1+zp|} &= \frac{|1/z|}{|\frac{1}{z}+p|} \\ &\leq \frac{|1/z|}{|\text{Im } \frac{1}{z}|} \\ &= \frac{|\overline{z}|/|z|^2}{|\text{Im } \overline{z}|/|z|^2} \\ &= \frac{|z|}{|\text{Im } z|}\end{aligned}$$

which gives the bound

$$|g(z)| \leq \int_{-\infty}^{\infty} \frac{|z|}{|\text{Im } z|} d\mu(p) = \frac{|z|}{|\text{Im } z|} c_0 \tag{1.6.1}$$

Since the integrand has a singularity only when $p = -1/z$ the Stieltjes transform of a measure of bounded support must converge in some neighborhood of zero. This cannot be said in general for Stieltjes transforms.

1.6.2 Remark. The bound above has a helpful geometric interpretation. For starters we note that,

$$\frac{|\text{Im } z|}{|z|} = |\sin(\arg z)|,$$

which can be seen by considering the right triangle formed by the points 0, z, and Re z. Thus this bound essentially depends on the angle of separation between z and the real line.

1.6.3 Proposition. The function $g(z)$ is analytic on the upper and lower half planes Im $z \neq 0$.

Proof. Let z, w be complex numbers with non-zero imaginary part. We have

$$\begin{aligned} g(w) - g(z) &= \int_{-\infty}^{\infty} \frac{1}{1+wp} - \frac{1}{1+zp} d\mu(p) \\ &= \int_{-\infty}^{\infty} \frac{(1+zp)-(1+wp)}{(1+wp)(1+zp)} d\mu(p) \\ &= (w-z) \int_{-\infty}^{\infty} \frac{-p}{(1+wp)(1+zp)} d\mu(p). \end{aligned}$$

Thus it seems

$$\lim_{w \to z} \frac{g(w)-g(z)}{w-z} = \int_{-\infty}^{\infty} \frac{-p}{(1+zp)^2} d\mu(p).$$

To be precise we may justify the interchange of limit and integral here by dominated convergence. Recall that it suffices to find an integrable function $h : \mathbb{R} \to \mathbb{C}$ such that, for any w in a neighborhood of z,

$$\left| \frac{-p}{(1+wp)(1+zp)} \right| \leq h(p) \qquad \text{for all } p \in \mathbb{R}.$$

It should be noted that our particular dominating function $h(p)$ is somewhat arbitrary, and not a sharp bound. By the previously discussed bound we have,

$$\left| \frac{-p}{(1+wp)(1+zp)} \right| \leq |p| \frac{|wz|}{|\text{Im } w \text{ Im } z|}.$$

Furthermore $|w|/|\text{Im } w|$ is clearly continuous on the half planes, so in some neighborhood of

$$|p|\frac{|wz|}{|\text{Im } w \text{ Im } z|} \leq |p|\left(\frac{|z|^2}{|\text{Im } z|^2} + \epsilon\right)$$

for an arbitrary $\epsilon > 0$. Finally, we note that the right hand dominating function above is integrable since μ has finite moments:

$$\int_{-\infty}^{\infty} |p| d\mu \leq \int_{-\infty}^{\infty} \frac{1}{2}(p^2 + 1)\ d\mu = \frac{c_2 + c_0}{2} < \infty$$

$\square$

1.6.4 Remark. Since μ is a real measure, $g(z)$ commutes with conjugation,

$$g(\bar{z}) = \int_{\infty}^{\infty} \frac{d\mu}{1 + \bar{z}p} = \overline{\int_{\infty}^{\infty} \frac{d\mu}{1 + zp}} = \overline{g(z)}.$$

For this reason, although $g(z)$ is defined for all non-real z, we will only concern ourselves with its values on the upper half plane Im $z > 0$ going forward. As a consequence in chapter 3 we restrict sample points to the upper half plane to avoid redundancy.

When z is a real number $g(z)$ may not converge. In particular $g(z)$ does not converge when $z = -1/p$ for some p in the support of μ. Thus if μ has bounded support g converges in a neighborhood of 0. On the other hand if μ has unbounded support $g(0)$ is not defined. However — as we will see — even in the Hamburger transform case, a formal series expansion at $z = 0$ holds valuable information. The results of this section are described in detail in section 5.6 of [1].

By a geometric series expansion,

$$\frac{1}{1+zp} = \sum_{k=0}^{\infty} (-z)^k p^k$$

when $|zp| < 1$. This suggests the connection between $g(z)$ and the moment sequence $(c_k)_{k\in\mathbb{N}_0}$. Indeed, at least formally,

$$\begin{aligned} \int_{\infty}^{\infty} \frac{d\mu}{1+zp} &= \int_{\infty}^{\infty} \sum_{k=0}^{\infty} (-z)^k p^k \; d\mu \\ &= \sum_{k=0}^{\infty} (-z)^k \int_{\infty}^{\infty} p^k \; d\mu \\ &= \sum_{k=0}^{\infty} c_k (-z)^k. \end{aligned}$$

In the Markov case we have a positive radius of convergence. But even in the Hamburger case, there is a sense in which this asymptotic expansion holds "non-tangentially".

Non-tangential convergence at $z = 0$ is, as the name implies, convergence along curves not tangent to the real line at 0. More precisely, one defines a "wedge domain"

$$\Gamma_\delta = \{\delta \leq \arg z \leq \pi - \delta\}$$

in which curves approach 0 with an angle at least δ from the real line.

1.6.5 Definition. (Non-tangential convergence and asymptotic expansion) We

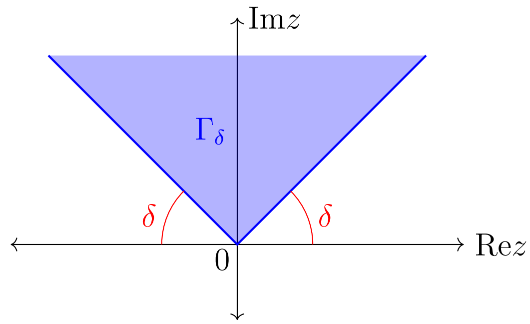

Figure 1.6.1: A wedge domain

say $h(z) \to 0$ non-tangentially as $z \to 0$ if, for any $\delta > 0$, the limit

$$\lim_{z \to 0} h(z) = 0$$

holds for z in the Γ_δ. If a formal power series $\sum_{k=0}^{\infty} a_k z^k$ and a function $g(z)$ are such that for all $n \in \mathbb{N}_0$,

$$g(z) = \sum_{k=0}^{n-1} c_k (-z)^k + z^n h_n(z)$$

where the trailing terms $h_n(z)$ converge to 0 non-tangentially as $z \to 0$, we say that $\sum_{k=0}^{\infty} a_k z^k$ is an asymptotic expansion of $g(z)$. We use the notation

$$g(z) \simeq \sum_{k=0}^{\infty} a_k z^k$$

for asymptotic expansions.

While $g(z)$ does not generally converge at $z = 0$, in a non-tangential sense the aforementioned asymptotic expansion holds:

1.6.6 Proposition. (i) If μ has bounded support

$$g(z) = \sum_{k=0}^{\infty} c_k (-z)^k$$

in a neighboorhood of $z = 0$.

(ii) For all $n \in \mathbb{N}_0$

$$g(z) = \sum_{k=0}^{n-1} c_k (-z)^k + z^n h_n(z) \tag{1.6.2}$$

where h_n is such that $\lim_{z \to 0} h_n(z) = 0$ non-tangentially.

Proof. We first note that

$$\sum_{k=0}^{n} c_k (-z)^k = \sum_{k=0}^{n} \int_{-\infty}^{\infty} (-zp)^k d\mu = \int_{-\infty}^{\infty} \frac{1 - (-zp)^{n+1}}{1 + zp} d\mu$$

and so

$$h_n(z) = \frac{1}{z^n} \left\{ g(z) - \sum_{k=0}^{n} c_k (-z)^k \right\} = \int_{-\infty}^{\infty} \frac{zp^{n+1}}{1 + zp} d\mu.$$

It remains to determine if and in what sense this trailing term $h_n(z)$ vanishes at the origin.

As expected in the case when μ has bounded support, $h_n(z)$ exists in a neighborhood of 0 and vanishes as $z \to 0$ in this neighborhood. Thus the asymptotic expansion 1.6.2 is the Taylor expansion: The Markov transform is analytic at the origin.

When μ has unbounded support and $h_n(z)$ may not be well defined on the real line, so we must weaken our result to non-tangential convergence. If $z \in \Gamma_\delta$ then we have the following inequalities for any $p \in \mathbb{R}$,

$$|p - z| \geq |p| \sin \delta \quad \text{and} \quad |p - z| \geq |p| \sin \delta.$$

We will show that

$$\lim_{z \to 0} h_{2n}(z) = 0, \quad z \in \Gamma_\delta$$

from which the corresponding limit for odd orders will immediately follow from the relation

$$h_{2n-1}(z) = zh_{2n}(z) + c_{2n} z^{2n}.$$

Here for the sake of convenience we define $w = -\frac{1}{z}$, following a more classical approach. Note that $|w| = \frac{1}{|z|}$ and if z is in the wedge Γ_δ then so is w. Now

$$\begin{aligned} |h_{2n}(z)| &\leq \int_{-\infty}^{\infty} \frac{|p|^{2n+1}}{|p - w|} d\mu \\ &\leq \frac{|z|}{\sin \delta} \int_{|p| \leq A} |p|^{2n+1} d\mu + \frac{1}{\sin \delta} \int_{|p| \geq A} p^{2n} d\mu \\ &\leq \frac{2|z| A^{2n+2}}{\sin \delta} + \frac{1}{\sin \delta} \int_{|p| \geq A} p^{2n} d\mu \end{aligned}$$

Thus

$$\lim_{z \to 0} |h_{2n}(z)| \leq \frac{1}{\sin \delta} \int_{|p| \geq A} p^{2n} d\mu, \quad z \in \Gamma_\delta.$$

Since A is arbitrary and the integral $\int p^{2n} d\mu = c_{2n}$ is convergent we are done. $\square$

Evidently the asymptotic series expansion $g(z) \simeq \sum_{k=0}^{\infty} c_k(-z)^k$ is generally formal, having zero radius of convergence except in the Markov case. However as we will see in the next section, rational functions constructed from this formal series exist which approximate $g(z)$ on the upper half plane Im $z > 0$.

Finally, we mention a relationship between the Stieltjes transform of an RT and the "multivariate Stieltjes" transform:

1.6.7 Definition. The **multivariate Stieltjes transform** of a Borel measure μ on $\mathbb{R}^n$ is defined by

$$g(y) = \int_{\mathbb{R}^n} \frac{d\mu(x)}{1 + \langle x, y \rangle} \simeq \sum_{\alpha \in \mathbb{N}_0^n} c_\alpha (-y)^\alpha$$

when μ has finite moments $c_\alpha = \int_{\mathbb{R}^n} x^\alpha d\mu$. The series expansion is generally formal, but again can be understood via the notion of non-tangential convergence.

1.6.8 Remark. Much like the univariate case, for measures μ supported on a bounded subset of $\mathbb{R}^n$, the multivariate Stieltjes transform is well defined in a neighborhood of the origin and equal to the power series expansion.

If μ has unbounded support in $\mathbb{R}^n$ we can not expect $g(y)$ to be defined on $\mathbb{R}^n$, and are forced to consider the Stieltjes transform as a function of multiple complex variables $g : \mathbb{C}^n \to \mathbb{C}$. Here we see singularities in the integrand when $y \in \mathbb{C}^n$

satisfies

$$\langle x, y \rangle = -1, \text{ for some } x \in \mathbb{R}^n$$

We should also note that we implicitly take the complex inner product here assumed to be conjugate linear in the first entry.

A precise description of the domain of definition for the multivariate Stieltjes transform will not be necessary for our purposes. Instead we will take a sufficient condition from the following slice relations:

1.6.9 Proposition. If $f \in L^1(\mathbb{R})$ has compact support then taking $F(p) = (1+zp)^{-1}$ in the slice theorem 1.4.2 we have

$$\int_{-\infty}^{\infty} \frac{R_f(\omega, p)}{1+zp}\, dp = \int_{\mathbb{R}^n} \frac{f(x)}{1+z\langle x, \omega \rangle}\, dx.$$

Similarly if $f \in L^1$ with possibly unbounded support by the Gaussian slice theorem 1.4.3,

$$\int_{-\infty}^{\infty} \frac{GR_f(\omega, p)\gamma(p)}{1+zp}\, dp = \int_{\mathbb{R}^n} \frac{f(x)\gamma_n(x)}{1+z\langle x, \omega \rangle}\, dx.$$

Note that the first relation makes sense in some neighborhood of $z = 0$, while the second applies only if $\operatorname{Im} z \neq 0$.

In chapter 3 we note that the integrals above can be seen as particular restrictions of the multivariate Stieltjes transform. That is, for $y \in \mathbb{C}^n$ of the form $y = z\omega$

where Im $z > 0$ and $\omega \in S^{n-1}$,

$$\int_{\mathbb{R}^n} \frac{f(x)}{1 + \langle x, y \rangle} \, dx = \int_{-\infty}^{\infty} \frac{R_f(\omega, p)}{1 + zp} \, dp.$$

Thus we consider the multivariate Stieltjes transform to be defined for $y \in \mathbb{C}^n$ in this form, as guarenteed by the relation above. An equivalent relation holds for the GRT via the previous proposition.

1.7 Padé approximants

In this section we recall the necessary definitions and results. Then we will prove some new results to be applied to the Gaussian Radon transform later.

1.7.1 Definition (Classical definition)**.** The Padé approximant to a (possibly formal) power series

$$R^{[L/M]}(z) \simeq \sum_{k=0}^{\infty} c_k z^k$$

is a rational function with numerator (denominator resp.) degree at most L (M resp.), with series equal to $\sum_{k=0}^{N} c_k z^k + O(z^{N+1})$ up to as high an order N as possible.

Let

$$R^{[L/M]}(z) = \frac{P^{[L/M]}(z)}{Q^{[L/M]}(z)} = \frac{a_L z^L + \cdots + a_1 z + a_0}{b_M z^M + \cdots + b_1 z + b_0}.$$

Notice that in general there is a negligible constant common factor between the numerator and denominator, so that with the remaining $L+M+1$ free parameters, we expect an order of accuracy of up to $L+M+1$ constraints, $c_0, c_1, \ldots, c_{L+M}$. Thus we define the $[L/M]$ Padé approximant by the condition,

$$\frac{P^{[L/M]}(z)}{Q^{[L/M]}(z)} = \sum_{k=0}^{L+M} c_k z^k + O\left(z^{L+M+1}\right). \tag{1.7.1}$$

Multiplying by $Q^{[L/M]}(z)$ gives a necessary condition, with a subtle but important difference from (1.7.1)

$$P^{[L/M]}(z) = Q^{[L/M]}(z)\left(\sum_{k=0}^{L+M} c_k z^k\right) + O\left(z^{L+M+1}\right). \tag{1.7.2}$$

In the classical theory of Padé approximants (1.7.2) was often taken as a definition. It is always possible to find polynomials of the required degree satisfying this second condition, however they do not necessarily attain the degree of accuracy required by the first. We follow Baker, defining $R^{[L/M]}(z)$ by (1.7.1), provided such a rational function exists.

1.7.2 Definition (Baker's definition)**.** The Padé approximant $R^{[L/M]}$ to a (possibly formal) power series is a the unique rational function with numerator (denominator resp.) degree at most L (M resp.) satisfying the condition (1.7.1)

$$\frac{P^{[L/M]}(z)}{Q^{[L/M]}(z)} = \sum_{k=0}^{L+M} c_k z^k + O\left(z^{L+M+1}\right).$$

If no such rational function exists we say the Padé approximant does not exist.

Note that the two definitions are equivalent if $b_0 = Q^{L/M}(0) \neq 0$. Hence a sufficient condition for the existence of the Padé approximant by Baker's definition is that $b_0 = Q^{L/M}(0) \neq 0$.

Equating coefficients of z^k, $k = 0, 1, \ldots, L+M$ in (1.7.2) gives two linear systems

$$
\begin{aligned}
a_0 &= b_0 c_0 \\
a_1 &= b_1 c_0 + b_0 c_1 \\
&\vdots \\
a_L &= b_L c_0 + b_{L-1} c_1 + \cdots + b_0 c_L
\end{aligned}
$$

and,

$$
\begin{aligned}
0 &= b_M c_{L-M+1} + b_{M-1} c_{L-M+2} + \cdots + b_0 c_{L+1} \\
0 &= b_M c_{L-M+2} + b_{M-1} c_{L-M+3} + \cdots + b_0 c_{L+2} \\
&\vdots \\
0 &= b_M c_L + b_{M-1} c_{L+1} + \cdots + b_0 c_{L+M}
\end{aligned}
$$

where for convenience we set $c_k = 0$ for $k < 0$. The first system shows the numerator $P^{[L/M]}$ is determined by the denominator $Q^{[L/M]}$ (explicitly computing the each coefficient $a_1, \ldots, a_L$ from $b_1, \ldots b_L$). The second linear system is homogeneous, with M equations in $M + 1$ unknowns. Thus we are guaranteed a nontrivial solution, and hence a Padé approximant by the classical definition.

A clever modification of the latter system gives us a determinantal formula for $Q^{[L/M]}$. Augmented with the desired definition,

$$Q^{[L/M]}(z) = b_M z^M + b_{M-1} z^{M-1} + \cdots + b_0$$

the system (now with dimension $M+1 \times M+1$) can be written in vector form,

$$\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ Q^{[L/M]}(z) \end{pmatrix} = \begin{pmatrix} c_{L-M+1} & c_{L-M+2} & \cdots & c_{L+1} \\ c_{L-M+2} & c_{L-M+3} & \cdots & c_{L+2} \\ \vdots & \vdots & \ddots & \vdots \\ c_L & c_{L+1} & \cdots & c_{L+M} \\ z^M & z^{M-1} & \cdots & 1 \end{pmatrix} \begin{pmatrix} b_M \\ b_{M-1} \\ \vdots \\ b_1 \\ b_0 \end{pmatrix}$$

Solving for b_0 by Cramer's Rule we get,

$$b_0 \begin{vmatrix} c_{L-M+1} & c_{L-M+2} & \cdots & c_{L+1} \\ c_{L-M+2} & c_{L-M+3} & \cdots & c_{L+2} \\ \vdots & \vdots & \ddots & \vdots \\ c_L & c_{L+1} & \cdots & c_{L+M} \\ z^M & z^{M-1} & \cdots & 1 \end{vmatrix} = \begin{vmatrix} c_{L-M+1} & c_{L-M+2} & \cdots & c_L \\ c_{L-M+2} & c_{L-M+3} & \cdots & c_{L+1} \\ \vdots & \vdots & \ddots & \vdots \\ c_L & c_{L+1} & \cdots & c_{L+M-1} \end{vmatrix} Q^{[L/M]}(z).$$

The LHS is a polynomial by Laplace expansion, and is nonzero if the RHS minor has nonzero determinant. Thus if the determinant

$$\begin{vmatrix} c_{L-M+1} & \cdots & c_L \\ \vdots & \ddots & \vdots \\ c_L & \cdots & c_{L+M-1} \end{vmatrix}$$

is nonzero, then up to the aforementioned constant factor,

$$Q^{[L/M]}(z) = \begin{vmatrix} c_{L-M+1} & \cdots & c_{L+1} \\ \vdots & \ddots & \vdots \\ c_L & \cdots & c_{L+M} \\ z^M & \cdots & 1 \end{vmatrix} \tag{1.7.3}$$

By convention we normalize $R^{[L/M]}(z)$ so that $b_0 = Q^{[L/M]}(0) = 1$. In terms of the determinental formula above, we can write $Q^{[L/M]}(0)$ as

$$Q^{[L/M]}(0) = \begin{vmatrix} c_{L-M+1} & \cdots & c_L & c_{L+1} \\ \vdots & \ddots & \vdots & \vdots \\ c_L & \cdots & c_{L+M-1} & c_{L+M} \\ 0^M & \cdots & 0 & 1 \end{vmatrix} = \begin{vmatrix} c_{L-M+1} & \cdots & c_L \\ \vdots & \ddots & \vdots \\ c_L & \cdots & c_{L+M-1} \end{vmatrix}$$

Matrices (determinants resp.) of this type — that is with antidiagonals all constant — is called a Hankel matrices (Hankel determinants resp.). Since the Hankel determinants give a useful form for expressing some conditions of interest we will define $H(n, m)$ to be the determinant of the $(m + 1) \times (m + 1)$ Hankel matrix with $c_n, \ldots, c_{n+2m}$ in the antidiagonals,

$$H(n,m) := \begin{vmatrix} c_n & \cdots & c_{n+m} \\ \vdots & \ddots & \vdots \\ c_{n+m} & \cdots & c_{n+2m} \end{vmatrix}.$$

We can relate the two previous equations by

$$Q^{[L/M]}(0) = H(L-M+1, M-1)$$

so that the sufficient condition for the existence of $R^{[L/M]}(z)$ is $H(L-M+1, M-1) \neq 0$.

We now narrow our focus to Padé approximants to a Hamburger transform. Let μ be a Borel measure on $\mathbb{R}$ with finite moments $c_k = \int p^k \, d\mu$.

With the shape reconstruction application in mind, we will now assume the measure μ is absolutely continuous with respect to the Lebesgue measure and denote it $f(x)dx$ where $f(x)$ is some non-negative Lebesgue measurable function of $\mathbb{R}$. Note that by this assumption, μ cannot be supported on a finite set. The additional restriction that μ has infinite support turns out to be important when discussing the determinacy of μ.

Recall that the Hamburger transform of μ,

$$g(z) := \int_{\mathbb{R}} \frac{d\mu}{1+pz}$$

has the asymptotic expansion, in the sense of non-tangential limits,

$$g(z) \simeq \sum_{k=0}^{\infty} c_k(-z^k).$$

We call this formal power series a Hamburger series.

1.7.3 Lemma. The Hamburger moments c_k satisfy the determinental condition $H(2n, m) \neq 0$ for $n, m = 0, 1, \ldots$. That is,

$$\begin{vmatrix} c_{2n} & \cdots & c_{2n+m} \\ \vdots & \ddots & \vdots \\ c_{2n+m} & \cdots & c_{2n+2m} \end{vmatrix} \neq 0$$

Proof. Consider the bilinear quadratic form given by

$$G(\mathbf{x}, \mathbf{y}) := \mathbf{x}^\top \begin{pmatrix} c_{2n} & \cdots & c_{2n+m} \\ \vdots & \ddots & \vdots \\ c_{2n+m} & \cdots & c_{2n+2m} \end{pmatrix} \mathbf{y}$$

If $\mathbf{x} = (x_0, x_1, \ldots, x_m)^\top$ then

$$\begin{aligned} G(\mathbf{x}, \mathbf{x}) &= \sum_{i,j=0}^{m} x_i x_j c_{2n+i+j} \\ &= \sum_{i,j=0}^{m} x_i x_j \int p^{2n+i+j} \, d\mu \\ &= \int \sum_{i,j=0}^{m} x_i x_j p^{2n+i+j} \, d\mu \\ &= \int p^{2n} \left(\sum_{k=0}^{m} x_k p^k \right)^2 d\mu \geq 0. \end{aligned}$$

Thus $G(\mathbf{x}, \mathbf{y})$ is positive semi-definite. Furthermore, equality holds

$$\int p^{2n} \left(\sum_{k=0}^{m} x_k p^k \right)^2 d\mu = 0$$

if and only if μ is supported entirely on the zeros of the polynomial $p^n \sum_{k=0}^{m} x_k p^k$. However by assumption μ has infinite support. So in fact G is strictly positive definite. We conclude, by Sylvester's criterion for example, that associated Hankel matrix has positive determinant. That is, $H(2n, m) > 0$. $\square$

The previous lemma guarantees the existence of certain Padé approximants, specifically those with $L - M + 1$ even.

1.7.4 Proposition. If J is odd then the Padé approximant $R^{[L/L+J]}(z)$ to the Hamburger series exists in Baker's sense.

Let $R_M(z)$ be the off-diagonal $[M/M+1]$ approximant to the Hamburger series of μ,

$$R_M(z) = \frac{P_M(z)}{Q_M(z)} \approx \sum_{k=0}^{\infty} c_k (-z)^k$$

where $c_k = \int p^k f(p) dp$.

The following theorem, paraphrased from Baker [1], is central to the shape reconstruction method discussed in Chapter 3.

1.7.5 Definition. A family of continuous functions $\mathscr{F}$ is called **normal** if every sequence in $\mathscr{F}$ contains a subsequence which converges **locally uniformly**, that is, uniformly on compact sets.

1.7.6 Proposition. The off-diagonal Padé approximants $R_M(z)$ to a solvable Hamburger series $\sum_{k=0}^{\infty} c_k(-z)^k$ form a normal family. The limit of a convergent subsequence of approximants must have an integral representation

$$\int_{-\infty}^{\infty} \frac{d\mu}{1+zp}$$

where μ is some Borel measure with moments c_k.

1.7.7 Corollary. If the moments c_k are determinate, then the off-diagonal approximants to the Hamburger series converge locally uniformly to the Hamburger transform of μ.

We discuss sufficient conditions for determinacy in section 1.2 and chapter 2.

Finally it should be noted that the shape reconstruction method of [3] utilized a certain multivariate generalization of the Padé approximants defined in [4]. Since a defining feature of these multivariate Padé approximants is that they coincide with Baker's univariate approximants on slices — precisely corresponding to our method of relating the multivariate Stieltjes transform to the univariate Stieltjes transform of projections — we choose to skip this description.

1.8 Symmetric tensors and polynomials

In chapter 4 we make some limited use of a tensor notation for multivariable polynomials. For reference we record some basic concepts in this section.

Let V be a finite dimensional real vector space with basis $\{e_1, e_2, \ldots, e_n\}$. The tensor power $V^{\otimes k}$ is a real vector space, with basis

$$\{e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_k}\}_{(i_1, \ldots i_k) \in \{1, \ldots n\}^k}$$

By convention we order this basis lexicographically. For example in the case $n = 3$, $k = 2$,

$$\begin{aligned} \{e_1 \otimes e_1,\ e_1 \otimes e_2,\ e_1 \otimes e_3, \\ e_2 \otimes e_1,\ e_2 \otimes e_2,\ e_2 \otimes e_3, \\ e_3 \otimes e_1,\ e_3 \otimes e_2,\ e_3 \otimes e_3, \} \end{aligned}$$

For convenience we will use the vector notation $\mathbf{e} = (e_1, \ldots, e_n)$ for the basis of V and

$$\mathbf{e}_{\otimes i} = e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_k}$$

for the elements of the induced basis of $V^{\otimes k}$, where $i = (i_1, i_2, \ldots i_k) \in \{1, 2, \ldots, n\}^k$. There is a natural embedding $V^k \hookrightarrow V^{\otimes k}$ given by

$$(v_1, v_2, \ldots, v_k) \mapsto v_1 \otimes v_2 \otimes \cdots \otimes v_k$$

and the tensor power satisfies the following universal property: Any "multilinear" map out of V^k can be factored as a linear map out of $V^{\otimes k}$ composed with the inclusion above.

$$\begin{tikzcd} V^k \arrow[r, hook] \arrow[dr] & V^{\otimes k} \arrow[d] \\ & W \end{tikzcd}$$

Given a linear transformation $T : V \to V$, the tensor power $T^{\otimes k}$ is the unique linear map $V^{\otimes k} \to V^{\otimes k}$ such that

$$T^{\otimes k}(\mathbf{e}_{\otimes i}) = (Te_{i_1}) \otimes \cdots \otimes (Te_{i_k}).$$

In the basis $\{\mathbf{e}_{\otimes i}\}$ ordered lexicographically the matrix representation for $T^{\otimes k}$ is given explicitly by the k-fold Kronecker product of the matrix for T in the basis $\mathbf{e}$. Tensor powers of linear maps commute with transposes

$$(T^{\otimes k})^\top = (T^\top)^{\otimes k}$$

If V has an inner product $\langle \cdot, \cdot \rangle_V$ then an inner product on $V^{\otimes k}$ can be defined by

$$\langle \mathbf{e}_{\otimes i}, \mathbf{e}_{\otimes i'} \rangle_{V^{\otimes k}} = \left\langle e_{i_1}, e_{i'_1} \right\rangle_V \cdots \left\langle e_{i_k}, e_{i'_k} \right\rangle_V .$$

We will generally omit the subscripts and write $\langle \cdot, \cdot \rangle$ for either inner product, determined from context. For any linear transformation $T : V \to V$ we have

$$\left\langle T^{\otimes k}(\mathbf{e}_{\otimes i}), \mathbf{e}_{\otimes i'} \right\rangle = \left\langle \mathbf{e}_{\otimes i}, (T^\top)^{\otimes k}(\mathbf{e}_{\otimes i'}) \right\rangle .$$

In particular for an orthogonal transformation $R \in O(n)$ on $\mathbb{R}^n$ we have

$$\left\langle R^{\otimes k}(\mathbf{e}_{\otimes i}), \mathbf{e}_{\otimes i'} \right\rangle = \left\langle \mathbf{e}_{\otimes i}, (R^{-1})^{\otimes k}(\mathbf{e}_{\otimes i'}) \right\rangle$$

and for a projection P,

$$\left\langle P^{\otimes k}\mathbf{e}_{\otimes i}, \mathbf{e}_{\otimes i'} \right\rangle = \left\langle \mathbf{e}_{\otimes i}, P^{\otimes k}\mathbf{e}_{\otimes i'} \right\rangle .$$

Note that

$$\left\langle T^{\otimes k}(\mathbf{e}_{\otimes i}), \mathbf{e}_{\otimes i'} \right\rangle = \left\langle Te_{i_1}, e_{i'_1} \right\rangle \cdots \left\langle Te_{i_k}, e_{i'_k} \right\rangle = T_{i_1, i'_1} \cdots T_{i_k, i'_k}$$

is a product of entries in the matrix representation for T with basis $\mathbf{e}$.

The inner product allows for a convenient expression of polynomials. Henceforth we take $V = \mathbb{R}^n$. For $x \in \mathbb{R}^n$ a multi index $\alpha \in \mathbb{N}_0^n$ of degree $|\alpha| = k$ we have

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} = \prod_{i=1}^{n} \langle x, e_i \rangle^{\alpha_n} = \left\langle x^{\otimes k}, \mathbf{e}^{\otimes \alpha} \right\rangle$$

where we write

$$x^{\otimes k} = \underbrace{x \otimes x \otimes \cdots \otimes x}_{k \text{ times}}$$

and

$$\begin{aligned}\mathbf{e}^{\otimes\alpha} &= e_1^{\otimes\alpha_1} \otimes \cdots \otimes e_n^{\otimes\alpha_n} \\ &= \underbrace{e_1 \otimes \cdots \otimes e_1}_{\alpha_1 \text{ times}} \otimes \cdots \otimes \underbrace{e_n \otimes \cdots \otimes e_n}_{\alpha_n \text{ times}}\end{aligned}$$

For example, in $\mathbb{R}^3$ the monomial $x^{(1,1,2)} = x_1x_2x_3^2$ can be written

$$x_1x_2x_3^2 = \langle x \otimes x \otimes x \otimes x, e_1 \otimes e_2 \otimes e_3 \otimes e_3 \rangle$$

Now this monomial representation is not unique. Indeed, since the tensor inner product is given as a product of euclidean products, we may as well imagine $\mathbf{e}^{\otimes\alpha}$ to be commutative. More precisely, for any permutation $\sigma \in S_k$ we have equivalently,

$$\left\langle x^{\otimes k}, \mathbf{e}_{\otimes i} \right\rangle = \left\langle x^{\otimes k}, e_{\sigma^{-1}i_1} \otimes \cdots \otimes e_{\sigma^{-1}i_k} \right\rangle$$

for any $i \in \{1, \dots, n\}^k$. The basis $\{\mathbf{e}_{\otimes i}\}_{i \in \{1,\dots,n\}^k}$ can partitioned by equivalence under permutation to $\mathbf{e}^{\otimes\alpha}$ for each degree k multiindex α. Let

$$I(\alpha) = \{i \in \{1, \dots, n\}^k : \left\langle x^{\otimes k}, \mathbf{e}_{\otimes i} \right\rangle = \left\langle x^{\otimes k}, \mathbf{e}^{\otimes\alpha} \right\rangle\}$$

One can check that $\{I(\alpha)\}_{|\alpha|=k}$ is a partition of the k-tuples $\{1, \dots, n\}^k$. For example with $n = 2, k = 2$, we have

$$\begin{aligned} x_1^2 &= \langle x \otimes x, e_1 \otimes e_1 \rangle \\ x_1 x_2 &= \langle x \otimes x, e_1 \otimes e_2 \rangle = \langle x \otimes x, e_2 \otimes e_1 \rangle \\ x_2^2 &= \langle x \otimes x, e_2 \otimes e_2 \rangle \end{aligned}$$

Thus it seems more appropriate to write monomials in terms of "commutative" symmetric tensors. We define the symmetrization of a basis tensor $\mathbf{e}_{\otimes i}$ by averaging over permutations

$$P_s \mathbf{e}_{\otimes i} := \frac{1}{k!} \sum_{\sigma \in S_k} e_{\sigma i_1} \otimes \cdots \otimes e_{\sigma i_k}.$$

which can be seen as an orthogonal projection onto the subspace of symmetric tensors (tensors invariant under permutation).

The subspace of k-tensors invariant under permutation of indices is denoted $V^{\odot k} \subseteq V^{\otimes k}$. Elements may be defined by the projection,

$$v_1 \odot v_2 \odot \cdots \odot v_k = \frac{1}{k!} \sum_{\sigma \in S_k} v_{\sigma 1} \otimes v_{\sigma 2} \otimes \cdots \otimes v_{\sigma k}.$$

The non-decreasing tuples $i = (i_1, \ldots i_k) \in \{1, \ldots, n\}^k$ indexes a standard basis,

$$\mathbf{e}_{\odot i} := e_{i_1} \odot \cdots \odot e_{i_k}, \qquad 1 \leq i_1 \leq \cdots \leq i_k \leq n.$$

Even better, there is a one to one correspondence between non-decreasing k-tuples and multi-indeces of degree k,

$$\tilde{\alpha} = (\underbrace{1, \ldots, 1}_{\alpha_1 \text{ times}}, \underbrace{2, \ldots, 2}_{\alpha_2 \text{ times}}, \ldots, \underbrace{n, \ldots, n}_{\alpha_n \text{ times}})$$

so we may write the basis

$$\mathbf{e}^{\odot\alpha} := e_1^{\odot\alpha_1} \odot \cdots \odot e_n^{\odot\alpha_n}, \qquad \alpha \in \mathbb{N}_0^n, |\alpha| = k$$

The k-tensor inner product restricted to $V^{\odot k}$ says

$$\begin{aligned} \langle v_1 \odot \cdots \odot v_k, w_1 \odot \cdots \odot w_k \rangle &= \sum_{\sigma_1, \sigma_2 \in S_k} \prod \langle v_{\sigma_1 j}, w_{\sigma_2 j} \rangle \\ &= \sum_{\sigma \in S_k} \prod \langle v_{\sigma j}, w_j \rangle \\ &= \sum_{\sigma \in S_k} \prod \langle v, w_{\sigma j} \rangle . \end{aligned}$$

Now the monomial expression in terms of symmetric tensors takes the simple form

$$x^\alpha = \left\langle x^{\odot k}, \mathbf{e}^{\odot\alpha} \right\rangle .$$

Taken further, the correspondence between homogeneous polynomials of degree k and symmetric k-tensors can be shown to be a vector space isomorphism, and extended to an isomorphism between the entire polynomial space and the graded space formed by symmetric tensors of all orders.

Chapter 2

Moments of the RT and GRT

In this chapter we discuss some necessary conditions for the the determinacy of certain multivariate moment problems by way of the Radon and Gaussian Radon transforms.

2.1 Projection moments and Petersen's theorem

2.1.1 Proposition. Let $c_k(\omega) = \int_{-\infty}^{\infty} R_f(\omega,p)p^k dp$ be the projection moments of f in a fixed derection $\omega \in S^{n-1}$, where $f : \mathbb{R}^n \to [0,\infty)$ has finite multivariate moments c_α. Then

$$c_k(\omega) = \sum_{|\alpha|=k} \binom{k}{\alpha} \omega^\alpha c_\alpha$$

where $\binom{k}{\alpha} = \frac{k!}{\alpha_1!\alpha_2!\cdots\alpha_n!}$ are multinomial coefficients.

Proof. By the slice theorem (1.4.2) with $F(p) = p^k$,

$$\int_{-\infty}^{\infty} R_f(\omega,p)p^k \; dp = \int_{\mathbb{R}^n} f(x)\langle x,\omega\rangle^k \; dx.$$

Now $\langle x, \omega \rangle^k = (x_1\omega_1 + \cdots + x_n\omega_n)^k$ has the multinomial expansion

$$\langle x, \omega \rangle^k = \sum_{|\alpha|=k} \binom{k}{\alpha} x^\alpha \omega^\alpha.$$

Thus after a bit of rearranging we get

$$\begin{aligned} \int_{\mathbb{R}^n} f(x)\langle x, \omega \rangle^k \, dx &= \int_{\mathbb{R}^n} f(x) \sum_{|\alpha|=k} \binom{k}{\alpha} x^\alpha \omega^\alpha \, dx \\ &= \sum_{|\alpha|=k} \binom{k}{\alpha} \omega^\alpha \int_{\mathbb{R}^n} f(x) x^\alpha \, dx, \end{aligned}$$

where the integrands are precisely the kth degree multivariate moments of f. $\square$

Similarly, moments of the GRT (Gaussian projection moments) can be expressed in terms of multivariate gaussian moments.

2.1.2 Proposition. Let $c_k^\gamma(\omega) = \int_{-\infty}^{\infty} GR_f(\omega, p)p^k\gamma(p)dp$ be the Gaussian moments of the GRT of f at a fixed ω, where $f : \mathbb{R}^n \to [0, \infty)$ has finite Gaussian multivariate moments $c_\alpha^\gamma = \int_{\mathbb{R}^n} f(x)\gamma_n(x)x^\alpha dx$. Then

$$c_k^\gamma(\omega) = \sum_{|\alpha|=k} \binom{k}{\alpha} \omega^\alpha c_\alpha^\gamma.$$

Proof. The proof follows as it did for the RT. This time we apply the GRT slice theorem (1.4.3) with $F(p) = p^k$,

$$\int_{-\infty}^{\infty} GR_f(\omega, p)p^k\gamma(p) \, dp = \int_{\mathbb{R}^n} f(x)\langle x, \omega \rangle^k \gamma_n(x) \, dx$$

Again we use the multinomial expansion of $\langle x, \omega \rangle^n$ and rearrange:

$$\int_{\mathbb{R}^n} f(x)\langle x, \omega \rangle^k \gamma_n(x)\ dx = \sum_{|\alpha|=k} \binom{k}{\alpha} \omega^\alpha \int_{\mathbb{R}^n} f(x)\gamma_n(x) x^\alpha dx.$$

Thus

$$c_k^\gamma(\omega) = \sum_{|\alpha|=k} \binom{k}{\alpha} \omega^\alpha c_\alpha^\gamma.$$

□

2.1.3 Example. Let $e_1, e_2, \ldots, e_n \in S^{n-1}$ be the the standard basis for $\mathbb{R}^n$,

$$(1, 0, \ldots, 0),\ (0, 1, \ldots, 0),\ \ \ldots,\ \ (0, 0, \ldots, 1)$$

Then $\langle x, e_i \rangle = x_i$ is the natural projection of $\mathbb{R}^n$ onto the e_i axis. The standard projection moments can be calculated as follows

$$\begin{aligned} c_k(e_i) &= \sum_{|\alpha|=k} \binom{k}{\alpha} e_i^\alpha c_\alpha \\ &= c_{ke_i} \end{aligned}$$

since $e_i^\alpha = 0$ unless $\alpha = ke_i$.

The following theorem, due to Petersen [10], gives a way us to reduce the question of determinacy for multivariate moment problems to the classical case.

2.1.4 Proposition (Petersen's theorem)**.** Let μ be a Borel measure with finite moments on $\mathbb{R}^n$, and $e_1, \ldots, e_n$ the standard basis for $\mathbb{R}^n$. If each $R_\mu^{e_1}, \ldots, R_\mu^{e_n}$ is determinate as a measure on $\mathbb{R}$, then μ is determinate.

Proof. An outline of the proof is as follows: Preliminarily, note that the solution set $[\mu]$ of Borel measures with equivalent moments to μ is convex, and μ is an extreme point in $[\mu]$ if and only if polynomials are dense in $L^1(\mathbb{R}^n, \mu)$ [16]. Thus it suffices to show that polynomials are dense in $L^1(\mathbb{R}^n, \mu')$ for any $\mu' \in [\mu]$.

The family of products of continuous functions of compact support $f(x) = \prod_{i=1}^n f_i(x_i)$ where each $f_i \in C_c(\mathbb{R})$, is dense in $L^1(\mu)$. Furthermore, since $R_\mu^{e_i}$, $i = 1, \ldots, n$ are determinate, polynomials are dense in each $L^2(\mathbb{R}, R_\mu^{e_i})$. Petersen constructs a series of polynomials $P_i : \mathbb{R} \to \mathbb{R}$ such that the polynomial product $P(x) = \prod_{i=1}^n P_i(x_i)$ arbitrarily close to f in $L^1(\mu)$. Thus polynomials are dense in $L^1(\mathbb{R}^n, \mu)$ and the moment problem is determinate. $\square$

2.1.5 Remark. Petersen's theorem is not basis dependent. Indeed, given an orthonormal basis $\tilde{e}_1, \ldots, \tilde{e}_n \in \mathbb{R}^n$, the orthogonal matrix R defined by $Re_i = \tilde{e}_i$, $i = 1, \ldots, n$, induces a one-to-one mapping between moment net c_α and the moment net $\tilde{c}_\alpha$ given by

$$\tilde{c}_\alpha = \int_{\mathbb{R}^n} f(x) \sum_{i=1}^n \langle x, \tilde{e}_i \rangle \ dx$$

2.1.6 Corollary. If μ is compactly supported Borel measure on $\mathbb{R}^n$, then the multivariate moment problem is determinate.

2.2 Determinate Gaussian projection moments

As discussed in section1.3, any positive function f in the weighted space $L^2(\gamma)$ has determinate Gaussian moments. By Petersen's theorem if $GR_f(e_i, \cdot)$ has determinate Gaussian moments for each $i = 1, \ldots, n$ then the function $f : \mathbb{R}^n \to [0, \infty)$ has determinate Gaussian moments. For the purposes of the shape reconstruction method we note that the GRT of an indicator function is bounded,

$$GR_f(\omega, p) = \int\limits_{\langle x,\omega\rangle = p} f(x)\gamma_{n-1}(x - p\omega)dx \leq \int\limits_{\langle x,\omega\rangle = p} \gamma_{n-1}(x - p\omega)dx = 1,$$

thus all measurable sets have determinate Gaussian moments. We make a stronger claim:

2.2.1 Theorem. If $f : \mathbb{R}^n \to [0, \infty)$ is measurable and

$$\int_{\mathbb{R}^n} |f(x)|^2 \gamma_n(x)\ dx < \infty,$$

then the Gaussian moments of $GR_f(\omega, p)$ given by

$$c_k^\gamma(\omega) = \int_{-\infty}^{\infty} GR_f(\omega, p)p^k \gamma(p)\ dp$$

are determinate for any $\omega \in S^{n-1}$, and as a consequence the Gaussian moments of f are determinate.

Proof. Fix $\omega \in S^{n-1}$. By the Cauchy-Shwarz inequality,

$$|GR_f(\omega, p)|^2 \leq \int\limits_{\langle \omega,x\rangle = p} |f(x)|^2 \gamma_{n-1}(x - p\omega)\ dx.$$

Thus by the Gaussian slice theorem

$$\begin{aligned}\int_{-\infty}^{\infty} |GR_f(\omega,p)|^2 \, \gamma(p)\; dp &\leq \int_{-\infty}^{\infty} \int_{\langle x,\omega\rangle = p} |f(x)|^2 \, \gamma_{n-1}(x - p\omega)\; dx\; \gamma(p)\; dp \\ &= \int_{\mathbb{R}^n} |f(x)|^2 \gamma_n(x)\; dx < \infty.\end{aligned}$$

Since $GR_f(\omega,p) \in L^2(\gamma(p))$ the determinacy of the Gaussian projection moments follows. By Petersen's theorem, the Gaussian moments of f are determinate. $\square$

Chapter 3

Shape reconstruction from moments

In this chapter we describe a method for shape reconstruction from moments and propose an extension of the method to unbounded regions. In the first section we summarize the method for bounded sets. This is the work of Cuyt et al. [3] which we extend in the following sections to unbounded shapes and positive functions of at most polynomial growth. The methods all each have the basic structure:

(i) From moments of f compute Padé approximants to its Stieltjes transform.

(ii) Choosing a set of nodes on which to approximate f, construct an appropriate cubature formula to the Stieltjes transform.

(iii) Solve — or approximate a solution to — the linear system given by equating the Padé approximants and cubature formula.

We will not discuss the technical aspects of the reconstruction, other than to point to the computation examples given in [3], which utilize cubature formulas compiled in [13]. We have independenly verified that the method produces reasonably accurate reconstructions. To be clear however, our aim is only to show how the results of chapter 2 fit in to the shape reconstruction method, and suggest an extension to the method.

3.1 Reconstruction of bounded shapes

Suppose $A \subseteq \mathbb{R}^n$ is a bounded measurable set with indicator function f. Then it has finite multivariate moments

$$c_\alpha = \int_A x^\alpha \, dx.$$

The projection moments for any $\omega \in S^{n-1}$, given by

$$c_k(\omega) = \sum_{|\alpha|=k} \binom{k}{\alpha} c_\alpha \omega^\alpha$$

are coefficients to the Stieltjes series

$$\sum_{k=0}^{\infty} c_k(\omega)(-z)^k,$$

which is converges in a neighborhood of $z = 0$ to the Stieltjes transform

$$g_\omega(z) = \int_{-\infty}^{\infty} \frac{R_A(\omega, p)}{1 + pz} \, dp = \int_A \frac{1}{1 + z \langle x, \omega \rangle} \, dx.$$

Note that since A is bounded, say by $\|x\| \leq r$ for all $x \in A$, then $RT_A(\omega, p)$ vanishes for $|p| > r$. Singularities in the integrand can only occur when $z = -1/p$ so we conclude $g_\omega(z)$ is well defined for $|z| < 1/r$.

Recall that since A is bounded, it has determinate multivariate moments and projection moments. By corollary 1.7.7, $g_\omega(z)$ is the uniform limit of the off-diagonal Padé approximants $R_M(z)$ to the Stieltjes series. The rate of convergence in this case is known, and can be found in [1].

We consider $g_\omega(z)$ as the restriction of the multivariate Stieltjes transform

$$g(y) = \int_A \frac{1}{1 + \langle x, y \rangle} dx, \qquad y \in \mathbb{R}^n$$

to a the linear span of $\omega \in S^{n-1}$. That is, for $z \in \mathbb{R}$,

$$g(z\omega) = \int_A \frac{1}{1 + z \langle x, \omega \rangle} dx = g_\omega(z).$$

Here $g(z\omega)$ is well defined for $|z| < 1/r$ so that the multivariate transform is well defined for $\|y\| < 1/r$. For any $y \in \mathbb{R}^n$, we can compute projection moments with $\omega = y / \|y\|$, from which we can approximate $g(y) = g_\omega(\|y\|)$ by way of the Padé approximants described above. Let us denote this approximation by $\tilde{g}(y)$.

Now suppose $g(y)$ can be simultaneously approximated by a cubature formula

$$\sum_{i=0}^{I} w_i(y) f(x_i) \approx \int_A \frac{1}{1 + \langle x, y \rangle} \, dx,$$

where $f(x) = 1_A(x)$. Consider the linear system,

$$\sum_{i=0}^{I} w_i(y_j) \tilde{f}_i = \tilde{g}(y_j), \qquad j = 0, \ldots, J$$

where $y_0, \dots, y_J \in \mathbb{R}^n$ are a set of distinct sample points with $y_i < 1/r$ and $J > I$. At least heuristically, we expect a solution to this system in the form of the values to approximate the values of the characteristic function

$$\tilde{f}_i \approx f(x_i), \qquad i = 0, \dots, I,$$

at the cubature nodes x_i. Thus, if for example, we use a cubature formula with nodes x_i on a finite integer lattice around A, we will have computed an approximate pixel image of A.

3.2 Reconstruction of unbounded shapes

Our proposed extension to the method above follows roughly the same trajectory, but applies to a much broader class of functions. We will start with the prime example of indicator functions of unbounded measurable sets, and discuss the generalization in the next section.

Let $A \subseteq \mathbb{R}^n$ be a measurable set with indicator function f. Then f necessarily has finite Gaussian moments,

$$c_\alpha^\gamma = \int_{\mathbb{R}^n} f(x) x^\alpha \gamma_n(x)\ dx.$$

The Gaussian projection moments for any $\omega \in S^{n-1}$ are given by

$$c_k^\gamma(\omega) = \sum_{|\alpha|=k} \binom{k}{\alpha} c_\alpha^\gamma \omega^\alpha.$$

Now the Stieltjes series

$$\sum_{k=0}^{\infty} c_k^{\gamma}(\omega)(-z)^k$$

is generally formal. Likewise, the Stieltjes transform

$$g_\omega^\gamma(z) := \int_{-\infty}^{\infty} \frac{GR_A(\omega,p)}{1+pz}\gamma(p)\ dp = \int_A \frac{1}{1+z\,\langle x,\omega\rangle}\gamma_n(x)\ dx$$

may be undefined on the real line, but is analytic on the half planes Im $z \neq 0$. In this case the Stieltjes series is an asymptotic expansion to the Stieltjes transform in the sense of nontangential convergence,

$$g_\omega^\gamma(z) \simeq \sum_{k=0}^{\infty} c_k^{\gamma}(\omega)(-z)^k.$$

For more details on the Stieltjes series and transform in the case of Hamburger moments see section 1.6.

Now we note that $GR_A(\omega,p)$ is a bounded function, so that the Gaussian projection moments are determinate. Thus, as discussed in section 1.7, the $[M-1/M]$ Padé approximants to the Stieltjes series converge locally uniformly on the half planes to the Stieltjes transform.

Once again, we view $g_\omega^\gamma(z)$ as a restriction of a multivariate Stieltjes transform. Following the lead of the previous method, we consider the function $g^\gamma : \mathbb{C}^n \to \mathbb{C}^n$

given by the integral representation

$$g^{\gamma}(y) = \int_A \frac{1}{1 + \langle x, y \rangle} \gamma_n(x)\ dx$$

whenever well defined. Here $\langle \cdot, \cdot \rangle$ is the standard inner product on $\mathbb{C}^n$. We have seen that $g^{\gamma}(y)$ exists when $y = z\omega$ for $\omega \in S^{n-1}$ and $z \in \mathbb{C}$ has nonzero imaginary part,

$$g^{\gamma}(z\omega) = \int_A \frac{1}{1 + z \langle x, y \rangle} \gamma_n(x)\ dx = g^{\gamma}_{\omega}(z)$$

(We assume the complex inner product is conjugate linear in the first variable and linear in the second for simplicity).

Again, suppose $g^{\gamma}(y)$ can be simultaneously approximated by a cubature formula

$$\sum_{i=0}^{I} w_i(y) f^{\gamma}(x_i) \approx \int_A \frac{1}{1 + \langle x, y \rangle} \gamma_n(x)\ dx,$$

where $f^{\gamma}(x) = 1_A(x)\gamma_n(x)$. Consider the linear system,

$$\sum_{i=0}^{I} w_i(y_j) \tilde{f}^{\gamma}_i = \tilde{g}^{\gamma}(y_j), \qquad j = 0, \ldots, J$$

where $y_0, \ldots, y_J \in \mathbb{C}^n$ are a set of distinct sample points and $J > I$. In this case we require our sample points to take the form $y = z\omega$ sufficient for the existence of $g^{\gamma}(y)$. At least heuristically, we expect a solution to this system in the form of the values to approximate the values

$$\tilde{f}_i^\gamma \approx f^\gamma(x_i), \qquad i = 0, \dots, I,$$

at the cubature nodes x_i. In order to approximate A we simply remove the Gaussian weight,

$$\frac{\tilde{f}_i^\gamma}{\gamma_n(x_i)} \approx \frac{f^\gamma(x_i)}{\gamma_n(x_i)} = 1_A(x_i).$$

Here we may still use a finite lattice to produce a pixel image, but importantly neither the sample points y_i nor the Padé approximant $\tilde{g}^\gamma$ depend on the choice of cubature formula. In other words, with $\tilde{g}^\gamma(y_j)$ precomputed, image approximations to A can — in theory — be constructed for arbitrary pixel grids. We suspect that careful choice of cubature formulas may lead to additional redundancy in computing the weights $w_i(y_j)$.

3.3 Reconstruction of functions in weighted L^2 space

As discussed in section 2.2, the Gaussian moments of a function in the weighted space $L^2(\gamma_n)$ are determinate. Thus the method for unbounded shape reconstruction evidently applies just as well to all measurable functions $f : \mathbb{R}^n \to [0, \infty)$ which satisfy

$$\int_{\mathbb{R}^n} f(x)^2 \gamma_n(x)\, dx < \infty.$$

We will not repeat the construction, but in short, the Gaussian Moments

$$c_\alpha^\gamma = \int_{\mathbb{R}^n} f(x) x^\alpha \gamma_n(x)\ dx \qquad \alpha \in \mathbb{N}_0^n$$

can be used to approximate a multivariate Stieltjes transform

$$g(y) = \int_{\mathbb{R}^n} \frac{f(x)}{1 + \langle x, y \rangle} \gamma_n(x)\ dx$$

where $y \in \mathbb{C}^n$ is required to take a specific form. Simultaneously approximating $g(y)$ by a cubature formula, we forma a linear system for which approximate values of f on the cubature nodes is a solution.

Chapter 4

The Gaussian Radon transform of polynomials

Unlike the Radon transform, the Gaussian Radon transform is well defined on polynomials. In this chapter we investigate the GRT from this perspective, as a linear operator between polynomial spaces. In particular we prove a simple formula for the GRT of multivariate Hermite polynomials. We extend this result to some generalizations of the GRT.

We begin this chapter by proving a formula for the GRT of the multivariate Hermite polynomials.

4.1 The GRT of multivariate Hermite polynomials

Recall that for $\alpha \in \mathbb{N}_0^n$ the polynomials $H_\alpha(x)$ can be defined by the generating function

$$e^{\langle x,y\rangle - \frac{\|y\|^2}{2}} = \sum_{\alpha \in \mathbb{N}_0^n} \frac{H_\alpha(x)}{\alpha!} y^\alpha$$

where $\alpha! = \alpha_1! \cdots \alpha_n!$. These are related to the classical Hermite polynomials by

$$H_\alpha(x) = \prod_{i=1}^{n} H_{\alpha_i}(x_i)$$

where, for $k \in \mathbb{N}_0^\infty$, the classical polynomials $H_k(p)$ can be defined by

$$e^{pt-\frac{t^2}{2}} = \sum_{k=0}^{\infty} \frac{H_k(p)}{k!} t^k.$$

4.1.1 Proposition. Let $\omega \in S^{n-1}$ and $p \in \mathbb{R}$ be fixed. If $|\alpha| = k$, then the GRT of the multivariate Hermite polynomial H_α is

$$GR_{H_\alpha}(\omega, p) = H_k(p)\omega^\alpha \tag{4.1.1}$$

Proof. Recall that the generating function $\phi(x) = e^{\langle x,y\rangle - \frac{\|y\|^2}{2}}$ converges absolutely for all $x, y \in \mathbb{R}^n$. Consider the GRT of ϕ,

$$GR_\phi(\omega, p) = \int\limits_{\langle x,\omega\rangle = p} \phi(x)\gamma_{n-1}(x - p\omega)\ dx.$$

Immediately we can expand $\phi(x)$, interchanging integral and series, to see

$$\begin{aligned} GR_\phi(\omega, p) &= \sum_{\alpha\in\mathbb{N}_0^n} \frac{1}{\alpha!} y^\alpha \int\limits_{\langle x,\omega\rangle = p} H_\alpha(x)\gamma_{n-1}(x - p\omega)\ dx \\ &= \sum_{\alpha\in\mathbb{N}_0^n} \frac{GR_{H_\alpha}(\omega, p)}{\alpha!} y^\alpha. \end{aligned} \tag{4.1.2}$$

On the other hand, we will be able to show that

$$GR_\phi(\omega, p) = \sum_{k=0}^{\infty} \sum_{|\alpha|=k} \frac{H_k(p)\omega^\alpha}{\alpha!} y^\alpha. \tag{4.1.3}$$

Thus, being careful to note that the series (4.1.2) and (4.1.3) converge absolutely, we can compare coefficients for the desired formula. In order to derive the second expansion we begin by translating our integral onto the linear subspace $\langle x,\omega\rangle = 0$, via the change of variables $x \mapsto x + p\omega$:

$$\begin{aligned} GR_\phi(\omega,p) &= \int\limits_{\langle x,\omega\rangle=0} \phi(x+p\omega)\gamma_{n-1}(x)\ dx \\ &= \int\limits_{\langle x,\omega\rangle=0} e^{\langle x+p\omega,y\rangle-\frac{\|y\|^2}{2}}(2\pi)^{-\frac{n-1}{2}}e^{-\frac{\|x\|^2}{2}}\ dx \\ &= e^{p\langle\omega,y\rangle-\frac{\|y\|^2}{2}}\int\limits_{\langle x,\omega\rangle=0} e^{\langle x,y\rangle-\frac{\|x\|^2}{2}}(2\pi)^{-\frac{n-1}{2}}\ dx \end{aligned} \tag{4.1.4}$$

Now in order to compute the right side integral we consider the orthogonal decomposition $y = y_\omega + y_{\omega^\perp}$, where

$$y_\omega = \langle y,\omega\rangle\omega \qquad y_{\omega^\perp} = y - y_\omega.$$

We make two observations about this decomposition of y: First,

$$\|y\|^2 = \|y_\omega\|^2 + \|y_{\omega^\perp}\|^2 = \langle y,\omega\rangle^2 + \|y_{\omega^\perp}\|^2 \tag{4.1.5}$$

and second, for x in the linear subspace $\langle x,\omega\rangle = 0$ we have

$$\langle x,y\rangle = \langle x,\omega\rangle\langle y,\omega\rangle + \langle x,y_{\omega^\perp}\rangle = \langle x,y_{\omega^\perp}\rangle.$$

This second observation allows us to solve the integral at the end of (4.1.4). By completing the square $\|x - y_{\omega^\perp}\| = \|y_{\omega^\perp}\|^2 - 2\langle x, y_{\omega^\perp}\rangle + \|x\|^2$, we have

$$\begin{aligned}\int\limits_{\langle x,\omega\rangle=0} e^{\langle x,y\rangle - \frac{\|x\|^2}{2}} (2\pi)^{-\frac{n-1}{2}}\ dx &= e^{-\frac{\|y_{\omega^\perp}\|^2}{2}} \int\limits_{\langle x,\omega\rangle=0} e^{-\frac{\|y_{\omega^\perp}\|^2}{2} + \langle x, y_{\omega^\perp}\rangle - \frac{\|x\|^2}{2}} (2\pi)^{-\frac{n-1}{2}}\ dx \\ &= e^{-\frac{\|y_{\omega^\perp}\|^2}{2}} \int\limits_{\langle x,\omega\rangle=0} e^{-\frac{\|x-y_{\omega^\perp}\|}{2}} (2\pi)^{-\frac{n-1}{2}}\ dx\end{aligned}$$

which is just $e^{-\frac{\|p_{\omega^\perp}\|^2}{2}}$. It was important here to invoke the orthogonal projection $y_{\omega^\perp}$ so that the integrand becomes the translation of a standard Gaussian function on the hyperplane $\langle x, \omega\rangle = 0$. Now returning to (4.1.4), we have

$$\begin{aligned}GR_\phi(\omega, p) &= e^{p\langle \omega, y\rangle - \frac{\|y\|^2}{2} - \frac{\|y_{\omega^\perp}\|^2}{2}} \\ &= e^{p\langle \omega, y\rangle - \frac{\langle y,\omega\rangle^2}{2}}\end{aligned}$$

where we used (4.1.5). This looks like the generating function for the univariate Hermite polynomials $H_k(p)$ with $t = \langle \omega, y\rangle$, so we expand

$$GR_\phi(\omega, p) = \sum_{k=0}^{\infty} \frac{H_k(p)}{k!} \langle \omega, y\rangle^k.$$

Finally we apply the multinomial expansion of $\langle \omega, y\rangle^k$, recalling the multinomial coefficients $\binom{k}{\alpha} = \frac{k!}{\alpha!}$. Therefore

$$\begin{aligned}GR_\phi(\omega, p) &= \sum_{k=0}^{\infty} \sum_{|\alpha|=k} \frac{H_k(p)}{k!} \binom{k}{\alpha} y^\alpha \omega^\alpha \\ &= \sum_{k=0}^{\infty} \sum_{|\alpha|=k} \frac{H_k(p)\omega^\alpha}{\alpha!} y^\alpha\end{aligned}$$

as needed. □

4.1.2 Remark. The proof above was written by Dr. Sengupta. Recall that the Hermite polynomials H_α form a complete orthogonal basis for $L^2(\mathbb{R}^n, \gamma_n)$. In this sense the formula (4.1.1) completely defines the Gaussian Radon transform on this space. Indeed, it can be shown that any function $f \in L^2(\mathbb{R}^n, \gamma_n)$ has an expansion

$$f(x) = \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha^{(f)} H_\alpha(x)$$

where

$$a_\alpha^{(f)} = \int_{\mathbb{R}^n} f(x) H_\alpha(x) \gamma_n(x)\ dx.$$

If this expansion converges nicely on Λ then

$$\begin{aligned} GR_f(\Lambda) &= \int_\Lambda f(x)\gamma_n(x - p_0)\ dx \\ &= \int_\Lambda \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha^{(f)} H_\alpha(x)\gamma_n(x - p_0)\ dx \\ &= \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha^{(f)} \int_\Lambda H_\alpha(x)\gamma_n(x - p_0)\ dx \\ &= \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha^{(f)} GR_{H_\alpha}(\Lambda) \end{aligned}$$

If Λ is a hyperplane $\langle x, \omega \rangle = p$ with $\omega \in S^{n-1}$ then this gives

$$\begin{aligned} GR_f(\Lambda) &= \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha^{(f)} GR_{H_\alpha}(\Lambda) \\ &= \sum_{\alpha \in \mathbb{N}_0^n} a_\alpha^{(f)} H_{|\alpha|}(p) \omega^\alpha \\ &= \sum_{k=0}^{\infty} H_k(p) \sum_{|\alpha|=k} a_\alpha^{(f)} \omega^\alpha \end{aligned}$$

For details on the convergence of these expansions see [5] or [14].

4.2 The GRT of multivariate polynomials over general affine subspaces

In section 1.4 we defined the RT and GRT as the integrals of a function $f : \mathbb{R}^n \to \mathbb{R}$ over $n-1$ dimensional hyperplanes, the standard definitions. But there are many examples of geometric integral transforms, closely related to the hyperplane RT, which can be thought of as "generalized Radon transforms". In fact, the "Funk transform", which relates a function on the sphere S^3 to its integrals over great circles, was first introduced by Paul Funk in 1911, half a decade before Radon introduced the hyperplane RT. In the remainder of this chapter we discuss the notion of the RT and GRT on general d-dimensional affine subspaces of $\mathbb{R}^n$, where $d = 0, \ldots, n$. For context, note that when $d = 1$, this is the so called "X-ray transform" which gets it's name from the direct application to radiology.

Let Λ_0 be a d-dimensional linear subspace of $\mathbb{R}^n$, and $v \in \mathbb{R}^n$. Then the translation of Λ_0 by v

$$\Lambda = v + \Lambda_0$$

is called an **affine subspace** of $\mathbb{R}^n$. The representation above is clearly not unique since adding any member of Λ_0 to v does not change Λ. However, given an affine subspace Λ we have a canonical choice for v as the closest point in Λ to the origin. This point exists and is unique since Λ is closed and convex. Furthermore the point v defined in this way must be orthogonal to any vector in the linear subspace Λ_0. That is, $v \in \Lambda_0^\perp$, where $\Lambda_0^\perp$ is the linear subspace

$$\Lambda_0^\perp = \{x \in \mathbb{R}^n : \langle x, y \rangle = 0 \text{ for all } y \in \Lambda\}$$

also known as the **orthogonal complement** of Λ_0. Thus we rephrase our definition as follows:

4.2.1 Definition. An **affine subspace** Λ of dimension d in $\mathbb{R}^n$ is defined uniquely by

$$\Lambda = v + \Lambda_0$$

where Λ_0 is a linear subspace of dimension d and $v \in \Lambda_0^\perp$. Sometimes we call Λ a "d-plane".

4.2.2 Remark. Note that the hyperplanes in the original definition of the RT are $(n-1)$-planes, and the lines in the "X-ray transform" are 1-planes.

Recall that if Λ_0 is a fixed linear subspace of dimension d, the orthogonal complements $\Lambda_0^\perp$ is a linear subspace of dimension $n-d$. For any $x \in \mathbb{R}^n$ there is a unique orthogonal decomposition $x = x_{\Lambda_0} + x_{\Lambda_0^\perp}$ where $x_{\Lambda_0} \in \Lambda_0$ and $x_{\Lambda_0^\perp} \in \Lambda_0^\perp$. This idea can be summarized by the expression

$$\mathbb{R}^n = \Lambda_0 \oplus \Lambda_0^\perp.$$

Sometimes it is convenient to have an explicit coordinate system on an affine subspace. If $u^{(1)}, \ldots, u^{(d)}$ is a orthonormal basis for Λ_0 then there is an isometric embedding of $\mathbb{R}^d$ into $\mathbb{R}^n$ with image Λ given by $x(t) : \mathbb{R}^d \to \Lambda$ where

$$x(t) = v + t_1 u^{(1)} + \cdots + t_d u^{(d)}, \qquad t \in \mathbb{R}^d.$$

By this embedding can define the Euclidean measure dx on Λ as the pushforward of the Lebesgue measure on $\mathbb{R}^n$, and the standard Gaussian weight on Λ as $w_d(t)$. Note the $x(t)$ is defined in such a way that $x(0) = v$ which is essential when defining the Gaussian measure on Λ.

Other times we prefer not to invoke a basis-dependent isometry, and while $x(t)$ clearly depends on the choice of $u^1, \ldots, u^{(d)}$, we should note that both the Euclidean and Gaussian measures on Λ are invariant under orthogonal transformations about v and thus independent of basis.

Now we can define the natural analogue of the RT for affine subspaces, which is sometimes called the d-**plane transform**. To be consistent with the hyperplane

RT we will write $v = p\omega$, where $\omega \in S^{n-1} \cap \Lambda_o^{\perp}$ and $p \in \mathbb{R}$. Of course this notation is unique only up to the identification $(\omega, p) = (-\omega, -p)$.

4.2.3 Definition. Let $f : \mathbb{R}^n \to \mathbb{R}$. For any affine subspace $\Lambda = p\omega + \Lambda_0$, we define the **Radon Transform of f on** Λ by

$$R_f(\Lambda) = \int_{\Lambda} f(x)\ dx,$$

and the **Gaussian Radon Transform of f on** Λ by

$$GR_f(\Lambda) = \int_{\Lambda} f(x)\gamma_d(x - p\omega)\ dx,$$

where d is the dimension of the linear subspace Λ_0. In both definitions dx denotes the d-dimensional Lebesgue measure on Λ.

4.2.4 Remark. Explicitly, if $x(t) : \mathbb{R}^d \to \Lambda$ is a Euclidean isometry then

$$R_f(\Lambda) = \int_{\mathbb{R}^d} f(x(t))\ dt.$$

If furthermore $x(0) = p\omega$ then

$$GR_f(\Lambda) = \int_{\mathbb{R}^d} f(x(t))\gamma(t)\ dt.$$

However keep in mind that these definitions are independent of the particular isometry.

In order to generalize the formula (4.1.1) we begin with the integral of H_α over linear subspaces. Recall that the GRT of a function f over a linear subspace Λ_0 can be written

$$GR_f(\Lambda_0) = \int\limits_{\mathbb{R}^d \times \{0\}^{n-d}} f(Rx)\gamma_d(x)\, dx \tag{4.2.1}$$

where $R \in SO(n)$ is the a rotation sending $\mathbb{R}^d \subseteq \mathbb{R}^n$ to Λ_0. Thus we would like to investigate the action of rotations on certain functions of interest, namely monomials and Hermite polynomials.

4.2.5 Lemma. Let $f(x) = x^\alpha$ for some multi-index $\alpha \in \mathbb{N}_0^n$ of degree $|\alpha| = k$. Then the rotation $f(Rx)$ has the expansion

$$f(Rx) = \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta| = k}} c(R)^\alpha_\beta x^\beta. \tag{4.2.2}$$

The coefficients $c(R)^\alpha_\beta$ are given by

$$c(R)^\alpha_\beta = \sum_{i \in I(\beta)} \prod_{j=1}^n \left\langle Re_{i_j}, e_{\alpha^j} \right\rangle$$

where $\alpha^j = \ell$ if $0 < j - \alpha_1 - \cdots \alpha_{\ell-1} \leq \alpha_\ell$ and

$$I(\beta) = \{i \in \{1, \ldots, n\}^k : 1, \ldots, n \text{ have multiplicity } \beta_1, \ldots \beta_n \text{ in } i\}$$

Note that the inner products $\left\langle Re_{i_j}, e_{\alpha^j} \right\rangle$ are matrix entries for R.

Proof. First one should note that $(Rx)^\alpha$ is a homogeneous polynomial of degree k; a polynomial because R is a linear transformation, and homogeneous because

$R \in SO(n)$ so that for $\lambda \in \mathbb{R}$,

$$(R\lambda x)^\alpha = (\lambda R x)^\alpha = \lambda^k (R x^\alpha).$$

Using the tensor power notation from section 1.8, we can write

$$(Rx)^\alpha = \prod_{i=1}^{n} \langle Rx, e_i \rangle^{\alpha_i} = \left\langle R^{\otimes k} x^{\otimes k}, \mathbf{e}^{\otimes \alpha} \right\rangle = \left\langle x^{\otimes k}, (R^{-1})^{\otimes k} \mathbf{e}^{\otimes \alpha} \right\rangle$$

Now we expand over the basis $\{\mathbf{e}_{\otimes i}\}_{i \in \{1,\ldots,n\}^k}$, noting that each tuple $i \in \{1, \ldots, n\}^k$ is an anagram to exactly one multiindex of degree k. To be precise $\{I(\beta)\}_{|\beta|=k}$ is a partition of $\{1, \ldots, n\}^k$. Thus

$$\begin{aligned}
\left\langle x^{\otimes k}, (R^{-1})^{\otimes k} \mathbf{e}^{\otimes \alpha} \right\rangle &= \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta|=k}} \sum_{i \in I(\beta)} \left\langle x^{\otimes k}, \mathbf{e}_{\otimes i} \right\rangle \left\langle \mathbf{e}_{\otimes i}, (R^{-1})^{\otimes k} \mathbf{e}^{\otimes \alpha} \right\rangle \\
&= \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta|=k}} x^\beta \sum_{i \in I(\beta)} \left\langle \mathbf{e}_{\otimes i}, (R^{-1})^{\otimes k} \mathbf{e}^{\otimes \alpha} \right\rangle.
\end{aligned}$$

That we have the non-tensor expression

$$\left\langle \mathbf{e}_{\otimes i}, (R^{-1})^{\otimes k} \mathbf{e}^{\otimes \alpha} \right\rangle = \prod_{j=1}^{n} \left\langle e_{i_j}, R^{-1} e_{\alpha^j} \right\rangle = \prod_{j=1}^{n} \left\langle R e_{i_j}, e_{\alpha^j} \right\rangle$$

follows from the definition of the inner product on $(\mathbb{R}^n)^{\otimes k}$ and further notation from appendix A. $\square$

4.2.6 Remark. We could write the coefficient $c(R)^\alpha_\beta$ in a slightly different form.

If we define the non-decreasing tuple $\tilde{\alpha}$ associated with a multi-index α by

$$\tilde{\alpha} = (\underbrace{1, \ldots, 1}_{\alpha_1 \text{ times}}, \underbrace{2, \ldots, 2}_{\alpha_2 \text{ times}}, \ldots, \underbrace{n, \ldots, n}_{\alpha_n \text{ times}})$$

then

$$c(R)^{\alpha}_{\beta} = \sum_{\sigma \in S_k} \prod_{j=1}^{n} R_{\sigma \tilde{\beta}_j, \tilde{\alpha}_j}.$$

4.2.7 Example. For a simple example take the monomial $f(x) = x_1^2 x_2$. Here $n = 2$, $\alpha = (2, 1)$ and $k = 3$. The partition of $\{1, 2\}^3$ over multi-indices $|(\beta_1, \beta_2)| = 3$ is given by

$$I(3, 0) = \{(1, 1, 1)\}$$

$$I(2, 1) = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$$

$$I(1, 2) = \{(1, 2, 2), (2, 1, 2), (2, 2, 1)\}$$

$$I(0, 3) = \{(2, 2, 2)\}$$

Now note that $(\alpha^1, \alpha^2, \alpha^3) = (1, 1, 2)$. If $R \in SO(2)$ is a 2×2 rotation matrix with entries $R_{i,j} = \langle Re_i, e_j \rangle$ then the formula (4.2.2) says

$$\begin{aligned}
(Rx)^{\alpha} &= x_1^3 R_{1,1}^2 R_{1,2} \\
&\quad + x_1^2 x_2 \left(R_{1,1}^2 R_{2,2} + 2 R_{1,1} R_{2,1} R_{1,2} \right) \\
&\quad + x_1 x_2^2 \left(2 R_{1,1} R_{2,1} R_{2,2} + R_{2,1}^2 R_{1,2} \right) \\
&\quad + x_2^3 R_{2,1}^2 R_{2,2}
\end{aligned}$$

While tensor power notation is concise, this sort of representation is needed for computation.

4.2.8 Proposition. The GRT of a monomial $f(x) = x^\alpha$ on a linear subspace Λ_0 is given by

$$GR_f(\Lambda_0) = \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta| = k}} c(R)^\alpha_\beta c^G_\beta$$

where $c^G_\beta = \int_{\mathbb{R}^d} x^\beta \gamma_d(x)\, dx$ are the Gaussian moments on $\mathbb{R}^d$. **Should be indexed by only the first d entries of β? Feels weird**.

Proof. We take (4.2.1),

$$GR_f(\Lambda_0) = \int\limits_{\mathbb{R}^d \times \{0\}^{n-d}} (Rx)^\alpha \gamma_d(x)\, dx,$$

and expand by (4.2.2),

$$\int\limits_{\mathbb{R}^d \times \{0\}^{n-d}} (Rx)^\alpha \gamma_d(x)\, dx = \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta| = k}} c(R)^\alpha_\beta \int_{\mathbb{R}^d \times \{0\}^{n-d}} x^\beta \gamma_d(x)\, dx$$

as needed. □

4.2.9 Proposition. The GRT of a multivariate Hermite polynomial H_α on an affine subspace Λ is given by

$$GR_{H_\alpha}(\Lambda) = \alpha! \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta| = |\alpha|}} c(P_{\Lambda_0^\perp})^\beta_\alpha \frac{H_\beta(p_0)}{\beta!}$$

Proof. Once again we take the GRT of the generating function $\phi(x) = e^{\langle x,y\rangle - \frac{\|y\|^2}{2}}$, initially expanding

$$GR_\phi(\Lambda) = \sum_{\alpha \in \mathbb{N}_0^n} \frac{GR_{H_\alpha}(\Lambda)}{\alpha!} y^\alpha$$

Simultaneously we have

$$\begin{aligned} GR_\phi(\Lambda) &= e^{\langle p_0,y\rangle - \frac{\|y\|^2}{2}} (2\pi)^{-\frac{d}{2}} \int_\Lambda e^{\langle x,y\rangle - \frac{\|x\|^2}{2}}\ dx \\ &= e^{\langle p_0,y\rangle - \frac{\|y\|^2}{2} + \frac{\left\|P_{\Lambda_0} y\right\|^2}{2}} \\ &= e^{\langle p_0,y\rangle - \frac{\left\|P_{\Lambda_0^\perp} y\right\|^2}{2}} \\ &= \sum_{\alpha \in \mathbb{N}_0^n} \frac{H_\alpha(p_0)}{\alpha!} (P_{\Lambda_0^\perp} y)^\alpha \end{aligned}$$

Now we should have noted that the formula (4.2.2) applies not only to rotations but to any linear transformation, so we have

$$\begin{aligned} GR_\phi(\Lambda) &= \sum_{\alpha \in \mathbb{N}_0^n} \frac{H_\alpha(p_0)}{\alpha!} \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta| = |\alpha|}} c(P_{\Lambda_0^\perp})^\alpha_\beta y^\beta \\ &= \sum_{\beta \in \mathbb{N}_0^n} y^\beta \sum_{\substack{\alpha \in \mathbb{N}_0^n \\ |\alpha| = |\beta|}} c(P_{\Lambda_0^\perp})^\alpha_\beta \frac{H_\alpha(p_0)}{\alpha!} \end{aligned}$$

Comparing coefficients we conclude.

$$\frac{GR_{H_\alpha}(\Lambda)}{\alpha!} = \sum_{\substack{\beta \in \mathbb{N}_0^n \\ |\beta| = |\alpha|}} c(P_{\Lambda_0^\perp})^\beta_\alpha \frac{H_\beta(p_0)}{\beta!}$$

□

Bibliography

[1] George A. Baker and P. R. Graves-Morris, **Padé approximants**, Encyclopedia of Mathematics and Its Applications ; Section, Mathematics of Physics, no. v. 13-14, Addison-Wesley, Advanced Book Program, Reading, Mass, 1981.

[2] V. I. Bogachev, **Gaussian measures**, Mathematical surveys and monographs, no. v. 62, American Mathematical Society, Providence, R.I, 1998 (eng).

[3] Annie Cuyt, **Multidimensional integral inversion, with applications in shape reconstruction**, SIAM Journal on Scientific Computing **27** (2005), no. 3.

[4] ______, **Padè approximants for operators: Theory and applications'**, Springer-Verlag, 84.

[5] Charles F. Dunkl and Yuan Xu, **Orthogonal polynomials of several variables**, Encyclopedia of Mathematics and Its Applications, vol. 81, Cambridge University Press, Cambridge, 2001. MR 1827871

[6] S. Helgason, **The Radon transform**, Birkhäuser, 1965.

[7] Irina Holmes and Ambar N. Sengupta, **A Gaussian Radon transform for Banach spaces**, J. Funct. Anal. **263** (2012), no. 11, 3689–3706. MR 2984081

[8] ______, **The Gaussian Radon transform in classical Wiener space**, Commun. Stoch. Anal. **8** (2014), no. 2, 247–268. MR 3269847

[9] Vochita Mihai, **The Radon-Gauss transform**, Doctor of Philosophy, Louisiana State University and Agricultural and Mechanical College, May 2004.

[10] L. C. Petersen, **On the relation between the multidimensional moment problem and the one-dimensional moment problem**, Math. Scand. **51** (1982), no. 2, 361–366. MR 690537

[11] Johann Radon, **Über die Bestimmung von Funktionen durch ihre Integralwerte längs gewissermannigfaltigkeiten**, (1917).

[12] ———, **On the determination of functions from their integral values along certain manifolds**, (1986).

[13] A. H. Stroud, **Approximate calculation of multiple integrals**, Prentice-Hall, 1971.

[14] Sundaram Thangavelu, **Lectures on Hermite and Laguerre expansions**, Mathematical Notes, no. 42, Princeton University Press, Princeton, N.J, 1993.

[15] Lawrence Zalcman, **Uniqueness and nonuniqueness for the Radon transform**, Bull. London Math. Soc. **14** (1982), no. 3, 241–245. MR 656606

[16] Naum I. Achiezer, **The classical moment problem and some related questions in analysis**, Classics in Applied Mathematics, no. 82, Society for Industrial and Applied Mathematics, Philadelphia, Pennsylvania, 2021.